Simulating a Post-Automation Economy

An agent-based, stock-flow-consistent model of automation, microfounded wealth concentration, behavioural taxation, and capital mobility

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Abstract

We develop an agent-based, stock-flow-consistent model of an economy undergoing automation, built to ask which fiscal instrument reaches the durable surplus that artificial intelligence creates. The model separates two channels: a competitive return on reproducible robotic capital, and a mobile, foreign-held intellectual-property rent earned by AI. Production is an endogenous nested-CES technology; wealth concentration is microfounded through heterogeneous, persistent returns to wealth; and taxation and capital mobility are modelled as behavioural responses. The central result is that the durable surplus is the foreign-held AI rent, a cross-border licence fee that corporate, robot, and compute or token taxes largely miss and that only a source-based levy (a digital-services-style tax or a withholding) reaches. The appropriate policy depends decisively on whether a country owns the automation or imports it: for a host that owns the rent the problem is domestic inequality, reached by progressive and wealth taxes; for a rent-importing host the problem is base erosion and a gradual transfer of capital ownership abroad, which a residence-based wealth tax cannot reach. We report conditional orderings, stress-tested with global (Sobol) sensitivity and a formal stability analysis, rather than point forecasts.

Keywords: automation; artificial intelligence; stock-flow consistent; agent-based macroeconomics; taxation; wealth distribution; digital services tax; foreign ownership; capital mobility.

JEL classification: C63; E25; H23; O33; D31; F23.

1 Overview

This report documents a refined model of an automating economy, in which artificial intelligence and robotics progressively shift income from labour to capital, the ownership of that capital may sit with households, the state, or abroad, and the state may fund a universal basic income (UBI) through taxation. Because the right policy turns on whether a country owns the automation or imports it, the analysis is run for both: a country that hosts the AI owners (the United States) and one that pays the rent abroad (the United Kingdom, or the European Union), which, as section 8 shows, need almost opposite responses.

The refined version (v3) was built specifically to answer the most damaging criticisms of the first-generation model: that its wealth tax was frictionless, that its concentration engine was an imposed kinetic kernel rather than an economic mechanism, and that its calibration was not disciplined. It replaces the kinetic kernel with microfounded heterogeneous returns, makes the wealth tax induce avoidance and capital flight, replaces the ad-hoc investment rule with a stable differential-saving closure, and adds a formal local-stability analysis. Two further hardening steps, a per-agent offshore-wealth metric and a global sensitivity analysis, sharpen the central finding and are reported in sections 6 and 9.

The headline conclusion is threefold, and is throughout a positive, closure-conditional comparison rather than an optimal-tax or welfare claim. First, stock taxes (a wealth tax, or a progressive wealth tax) compress the wealth Gini sharply, from about 0.63 under laissez-faire to about 0.15, while flow taxes (income tax plus UBI) leave it near the laissez-faire level. Second, solvency is a distinct axis that turns on the adequacy of taxation, not on ownership form: adequately-taxed regimes stay solvent and even accumulate a sovereign equity fund out of surpluses, whereas a welfare state promising a universal income on too thin a tax base is pushed onto an explosive r-greater-than-g debt path. Third, a behavioural wealth tax introduces only a modest wedge between measured and true inequality once fled capital is attributed back, and the distributional results are governed mainly by the primitive dispersion of returns to wealth rather than by the behavioural elasticities that are hardest to estimate (the global sensitivity analysis quantifies this, with the caveat that those elasticities' indices are themselves imprecisely estimated at the sample size used). All are stated carefully below, including a correction to a more dramatic but mis-calibrated earlier version of the composition result. They hold under this model's closure (output at potential, investment as the goods-market residual, equity valued at book) and should be read as numerically supported within it, not as general theorems. Throughout, these are conditional results that depend on the model's assumptions; those assumptions and limitations, and which of the conclusions are robust to them, are set out in full in Appendix A.

Reproducibility. All results come from Monte Carlo runs over 10 to 12 seeds. Two accounting invariants (the four sectors' deposits sum to zero; the sum of sector net worths equals the capital stock) hold to machine precision across the reported runs and tested configurations, and are asserted in a suite of 142 passing tests. Level quantities are reported as a ratio to per-capita output, which is scale-free.

1.1 The live policy debate this speaks to

This is not an abstract exercise. Through 2025 and 2026 a concrete version of the question moved into mainstream politics on two fronts. The first is displacement: Anthropic's Dario Amodei warned that AI could remove a large share of entry-level white-collar roles and push unemployment into double digits within a few years (Axios, 2025), and a run of layoffs at firms including Amazon, Meta and UPS, announced alongside record AI spending, revived a long-standing worry that fewer taxed workers mean a thinner fiscal base (EL País, 2025). The second is what, if anything, to tax in response, a question that goes back to the Gates and Phelps robot-tax proposals.

The 2026 proposals cluster around a levy on AI usage. Mark Cuban floated a federal per-token charge on providers; the Michigan Senate candidate Mallory McMorrow built a "Token Tax" into a worker-protection plan funding an AI Workforce Reinvestment Fund; Elizabeth Warren argued in Time for an excise on data-centre energy with the door left open to bolder measures; DuckDuckGo's Gabriel Weinberg proposed a ten per cent surcharge on token charges (roughly the employer payroll-tax rate) held in a lockbox for displaced workers; and Amodei himself mooted a three per cent revenue levy on model usage. The first-principles case behind them is that across the OECD a worker costs about a third of labour cost in tax, so when an agent does the same job the value re-appears as margin or capital gain that is taxed far less, eroding the base (a risk the IMF flagged in 2024).

The pushback is equally sharp. Dave Friedman's response to Cuban makes four arguments against a token base: a token is an internal accounting unit, not a unit of value, work or energy; the base is endogenous to the taxpayer, since providers write their own tokenizers and can shrink the count without changing the activity; per-token prices have fallen roughly two-hundred-fold a year, so a fixed rate is either confiscatory or quickly indexed to nothing; and a provider-level tax on domestic firms is in effect a subsidy for foreign inference. Palmer Luckey makes the territorial point bluntly, that such a tax mostly makes foreign models more attractive while building the apparatus to monitor AI usage, and others (Sinofsky, Petersen) reject the unit outright. The academic reviews are cautious rather than hostile: the IMF (2024) and Brookings both decline to single out AI, preferring broader capital or consumption bases, with Brookings leaning toward a consumption-side levy folded into VAT with business-to-business use exempted.

This paper takes no side on whether to tax AI. It asks the question underneath the slogans: in an economy where AI shifts income from labour to a capital-and-IP layer that can sit abroad, which tax base actually reaches the durable surplus, and who ends up owning the automated capital. The two-channel results (section 6) bear directly on the choice of base, on the territoriality objection, and on the displaced-worker funding question, and section 8 reads the model against these specific proposals. One note on perspective: most of the proposals, and most of the pushback, are framed for the United States, the home of the providers, whereas the model is framed for a host that imports the AI rent (read the United Kingdom). That difference turns out to matter for whether the territoriality objection bites.

A word on what kind of claim this is, to pre-empt the obvious objection that the direction is already known. The model does not claim to discover that an AI rent shifts income towards capital, or that a foreign-held rent leaks abroad: the rent is an *input* to the model, a calibrated assumption, not a *finding* from it. Its value lies in four things a verbal argument cannot supply. It puts magnitudes and a ranking on instruments that all sound plausible in prose, so that "tax the rent" becomes the sharper claim that a source levy reaches a durable surplus an ordinary corporate or robot tax misses almost entirely. It adjudicates the cases where the intuitions genuinely point both ways, as with the territorial objection, which turns out to bite for the provider's home and not for the rent's host. It tracks compounding, the slow accumulation of a reinvested rent into majority foreign ownership of the capital stock, which no static argument surfaces. And it enforces consistency, holding the stock-flow accounts and the factor-market closure together so that a policy cannot be credited with revenue it could not raise or a public balance sheet it could not sustain. The contribution is therefore the quantified ranking and the conditions under which it flips, not the sign of the first derivative, which is assumed.

Because that difference is the organising idea of the paper, we make it explicit by separating two regimes. In the first the AI is owned at home, the case of the United States, where providers such as OpenAI and Anthropic are resident: the rent stays in the country, and the policy question is how its concentration among domestic owners is shared. In the second the AI is owned abroad, the case of the United Kingdom with a United States or Chinese owner: the rent leaves as a licence fee, and the policy question is how, if at all, the host can reach it and slow the transfer of ownership. We take the domestic-owner case first, as the simpler benchmark with a familiar policy menu, then turn to the foreign-owner case, which is this paper's main contribution and occupies most of section 6. A single parameter, the foreign-owned share of the AI IP, moves the model between the two.

A second dimension, beyond where the rent goes, is how large it becomes, and it is central to how the results should be read. The experiments hold the rent at a fixed share of cognitive value added, a roughly constant eighth of output, which is deliberately the conservative case. Two forces are more likely to push it up than to hold it flat. As the technology becomes essential and the market concentrates, the owner's pricing power grows, so the markup need not stay fixed. And as AI-native production does whole jobs rather than assisting inside an otherwise-human firm, the AI cluster captures a rising share of total output rather than a fixed slice of one cluster, which is arguably already visible in software. When the rent rises with automation in this way the surplus roughly doubles to a quarter of output, or approaches half in the extreme, and the foreign-ownership drift accelerates from about seventy per cent of the capital stock toward almost the whole of it (section 6, Figure 26). The flat-rent results reported everywhere else are therefore a floor, not a central estimate. The policy stakes scale with the rent: the same instruments still reach it, but they capture far more, the ownership transfer they curb is larger and faster, and the cost of waiting is correspondingly higher (section 8).

2 Prior and related work

The model sits at the intersection of five literatures.

2.1 Automation and the factor distribution

Acemoglu and Restrepo (2022) model tasks allocated between labour and capital, with automation expanding the set of tasks performed by capital. The CES task block here is the reduced macro form of that mechanism. The closest analytic antecedent for the wealth dimension is Moll, Rachel and Restrepo (2022), who link automation to the personal income and wealth distributions through a return gap that generalises Piketty's $r - g$, with top-tail inequality governed by a random growth process. The refined model's concentration engine is built directly on that random-growth logic. We note the empirical debate the model must respect: Rognlie (2015) shows much of the measured rise in the capital share is housing and markup, not productive capital, and the recent AI-labour evidence is mixed on whether displacement is yet visible in aggregate; section 4 frames the automation scenario accordingly.

2.2 Heterogeneous returns to wealth

The refined concentration engine rests on the empirical regularity that returns to wealth are heterogeneous and persistent across households (Fagereng, Guiso, Malacrino and Pistaferri 2020). The resulting Pareto tail is the random-growth result of Benhabib, Bisin and Zhu, in which idiosyncratic persistent returns plus a demographic reset produce a stationary heavy-tailed wealth distribution. This replaces the econophysics kernel used in the first generation.

2.3 Behavioural responses to wealth taxation

The behavioural wealth tax is calibrated to the empirical literature on avoidance and mobility, and the two channels are kept distinct because they have different welfare and revenue implications. The first is base erosion: Brülhart and co-authors estimate a bunching elasticity of declared taxable wealth with respect to the net-of-tax rate of around 0.7 to 0.8. The model maps this to the avoidance_elasticity parameter through a constant-elasticity base factor, taxable_base = $(1 - \tau_{\mathrm{w}})^{\varepsilon_{\mathrm{a}}}$ with $\varepsilon_{\mathrm{a}} = 0.75$, so a higher rate mechanically shrinks the base it applies to (under-declaration, reclassification, legal sheltering) without any wealth physically leaving. The second is relocation: Jakobsen, Kleven, Kolsrud, Landais and Muñoz find international migration responses to wealth taxes, on the order of a two per cent reduction in the stock of top wealth-holders per percentage point of rate. The model maps this to migration_semi_elast = 0.02 as a target OFFSHORE share, target_offshore = $0.02 \times (\tau_{\mathrm{w}}$ in pp) of household equity, towards which the offshore stock adjusts partially each period, drawn disproportionately from the top of the distribution. The distinction matters: avoidance reduces measured revenue but the wealth stays in the domestic distribution, whereas migration removes wealth from the domestic base but the per-agent offshore account still attributes it to its owner, so the "true" Gini (section 6) counts it while the "measured" Gini does not. Treating the two separately is what lets experiment B decompose the behavioural response into an avoidance leg and a mobility leg.

2.4 Stock-flow-consistent and agent-based macro

The accounting backbone follows Godley and Lavoie (2007); the agent-based stock-flow-consistent benchmark is Caiani and co-authors (2016). The closest agent-based prior art is Carvalho and Di Guilmi (2020), which studies technological unemployment and inequality but locates inequality on the income and credit side rather than in the ownership of automated capital. The standard methodological critique of this class (over-parameterisation, stylised-fact-only validation) is addressed in section 5 through a Sobol sensitivity analysis and the stationary-baseline and stability checks.

2.5 Optimal taxation and open economy

Guerreiro, Rebelo and Teles (2022) study the optimal taxation of robots; the present model is positive rather than optimal-tax. The foreign-ownership dimension uses a rest-of-world balance-sheet sector, motivated by the scale of profit repatriation documented by Parnreiter, Steinwärder and Kolhoff (2024).

3 The model

3.1 The model at a glance

The diagram shows the five parties and the flows between them. The firm produces in two channels: a routine cluster (routine workers plus robotic capital) and a cognitive cluster (cognitive workers plus AI compute). Wages and dividends flow to domestic households; taxes flow to the state, which pays a UBI back; and a foreign owner abroad (read the United States or China) supplies the robots and AI compute, owns equity, and collects the AI IP rent as a licence fee that leaves the country. Each tax lever (in sienna) acts at a different point, which is the whole reason the instruments differ.

3.2 Sectors and invariants

Four sectors carry balance sheets: households (N agents), the firm or capital-owning sector, the government, and the rest of the world (RoW). Deposits are a liquid claim that sums to zero across

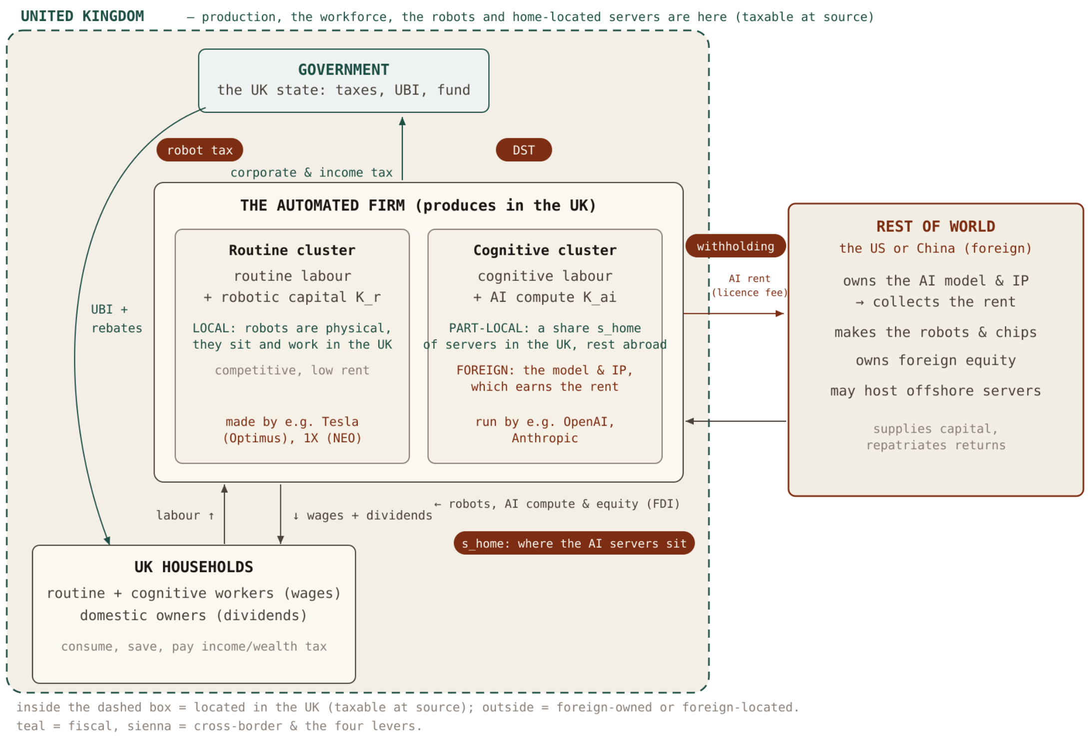

Figure 1a The parties and their relations, and what is local versus foreign. Everything inside the dashed boundary is located in the United Kingdom and is therefore reachable by tax at source: the workforce, the firm's operations, the physical robots, and the home-located share of the AI servers. Outside the boundary sits the foreign owner (read the United States or China), which owns the AI model and its intellectual property (the source of the rent), makes the robots and chips, owns foreign equity, and may host the offshore share of the servers. The routine cluster pairs routine workers with robotic capital that is fully physical and domestic; the cognitive cluster pairs cognitive workers with AI compute whose servers are split between home and abroad by the share s_home, while the model and IP behind it stay foreign. Grey flows are real and income flows (labour, wages and dividends between households and the firm); teal flows are fiscal (tax up to the state, UBI and rebates back to households); sienna flows cross the border (the AI rent leaving as a licence fee, and the inward supply of robots, compute and equity as foreign direct investment). The four sienna levers act at different points, which is the whole reason the instruments differ: the robot tax on the physical robotic income inside the UK, the DST on AI revenue earned in the UK, the withholding on the rent as it crosses the border, and s_home on how much of the AI compute is located, and so taxable, at home.

sectors; equity represents claims on the capital stock K, split between households, the state, and RoW. Two identities hold every period by construction and are tested to machine precision: the four sectors' deposit balances sum to zero, and the sum of sector net worths equals real assets (the capital stock K, with equity claims summing exactly to K and the firm holding no residual cash).

The two matrices below make the stock and flow structure inspectable in the stock-flow-consistent tradition (Godley and Lavoie 2007; Caiani et al. 2016). In the balance-sheet matrix every row sums to its economy-wide total (deposits to zero, equity to the capital stock once the firm's issued-equity liability is netted) and the net-worth row sums to the capital stock. In the transaction-flow matrix every row sums to zero (every payment is someone's receipt) and each column's current and capital entries balance, which is what the deposit and net-worth invariants verify numerically.

Balance-sheet matrix (end-of-period stocks; a plus sign is an asset, a minus sign a liability).

Stock	Households	Firm	Government	RoW	Σ
Capital		$+K$			$+K$
Deposits / bonds	$+D_h$	$+D_f \approx 0$	$+D_g$	$+D_r$	0
Equity (claims on K)	$+E_h$	$-K$	$+E_s$	$+E_r$	0
Net worth	E_h+D_h	≈ 0	E_s+D_g	E_r+D_r	$+K$

Deposits sum to zero (a government bond is a household/RoW asset and a government liability); equity claims sum to K, so once the firm's issued equity (−K) is netted against the capital it holds (+K) the firm's net worth is just its residual cash, which the closure keeps at zero. Household deposits and equity are both non-negative by construction (the no-overdraft rule and the equity floor), so household net worth is non-negative and the Gini needs no negative-value convention.

Transaction-flow matrix (within-period flows; a plus sign is a receipt/source, a minus sign a payment/use; the firm is split into a current and a capital account).

Flow	Households	Firm (cur.)	Firm (cap.)	Gov't	RoW	Σ
Consumption	$-C$	$+C$				0
Government spending		$+G$		$-G$		0
Wages	$+W$	$-W$				0
Profit (after tax)		$-\Pi$	$+\Pi$			0
Dividends	$+Div_h$		$-Div$	$+Div_s$	$+Div_r$	0
Corporate tax		$-T_c$		$+T_c$		0
Income + wealth tax	$-T_{y,w}$			$+T_{y,w}$		0
UBI	$+U$			$-U$		0
Interest on deposits	$+rD_h$			$+rD_g$	$+rD_r$	0
Investment			$-I$			$-I$
Δ Equity (issuance)	$-\Delta E_h$		$+\Delta E$	$-\Delta E_s$	$-\Delta E_r$	0
Δ Deposits	$-\Delta D_h$		$-\Delta D_f$	$-\Delta D_g$	$-\Delta D_r$	0

The investment row sits on the firm's capital account: gross saving (retained profit plus the equity savers subscribe) finances investment, so the capital-account column balances and the Δ-deposit row closes the period. Retained profit grows existing owners' claims pro-rata; new issuance is subscribed by whichever sectors are saving (households, the surplus government, RoW), which is how a persistent surplus is converted into an equity stake rather than an ever-growing deposit. The no-overdraft rule is the one place a household that cannot fund its allocation sells equity at book to a surplus sector, a matched swap that leaves both the Δ-equity and Δ-deposit rows summing to zero.

3.3 Production and factor prices

Output is a CES aggregate of a labour-task bundle and a capital-task bundle:

$$Y = A \cdot [\ (1 - I)\hat{}(1/e) \cdot (A\_L\ L)\hat{}((e-1)/e) + I\hat{}(1/e) \cdot (A\_K\ K)\hat{}((e-1)/e)\ ]\hat{}(e/(e-1))$$

The automation index I is the share of tasks performed by capital and e the elasticity of substitution. Factor prices are marginal products, so the capital share rises endogenously with automation, and the net return r and growth rate g are endogenous. With gross complements (e below one), potential output is bounded in capital. Output is supply-determined at potential, investment is the residual that clears the goods market ($S = I$), and consumption follows the differential-saving rule (high propensity out of labour income, low out of capital income, small wealth effect on equity). That rule pins the capital-output ratio at a stable interior value and removes the first-generation output blow-up, because scarcity rents on capital are saved and reinvested rather than consumed.

3.4 Concentration: heterogeneous persistent returns

Wealth concentrates because returns to equity are heterogeneous and persistent across households, an AR(1) return type per agent calibrated to the returns-to-wealth literature. A small demographic turnover (death and replacement by low-wealth entrants) pins a stationary Pareto tail. The additive inflow that prevents runaway condensation is the saving of wage earners who purchase equity. There is no imposed kinetic kernel and no artificial conservation: the tail is produced by the process, and the only rescaling enforces the balance-sheet identity that household claims sum to the household share of K.

The engine is disciplined against the standard cross-sectional calibration targets rather than only producing "a" heavy tail. The return primitives are set to the Fagereng et al. estimates: a cross-sectional standard deviation of household wealth returns of about five percentage points (ret_sigma = 0.05) and a high annual persistence of the return type (ret_persist = 0.92, in the range those panel studies report). With those primitives the laissez-faire benchmark settles at a wealth Gini of about 0.63, a top-1% share of about 24% and a top-10% share of about 53%, which bracket the empirical wealth concentration of advanced economies, and the upper tail follows an approximate power law whose slope corresponds to a Pareto exponent of roughly 1.5, consistent with the 1.4 to 1.7 range estimated for wealth. The return-dispersion comparative static (experiment C) is the explicit sensitivity to this calibration: SD of 0.03 / 0.05 / 0.07 maps to top-1% shares of about 12% / 24% / 39%, so the targets above are matched at the central value rather than by construction.

3.5 Behavioural policy and mobility

The wealth tax is levied on the stock and split into an equity leg (ownership transferred to the state) and a cash leg. It is no longer frictionless: the taxable base erodes with the rate (avoidance, calibrated to a bunching elasticity of 0.75), and domestically-held equity relocates abroad in response to the tax (capital flight, calibrated to a two per cent per percentage point semi-elasticity). The foreign ownership share is therefore an endogenous state variable that the tax itself moves. Other instruments (corporate and income tax, progressive wealth brackets, UBI, a citizens' wealth fund that rebates state dividends per capita) carry over from the first generation.

3.6 Baseline calibration

Parameter	Value	Meaning
N / periods	2000 / 600	agents and horizon
eps (e)	0.6	elasticity of substitution (gross complements, empirically defensible)
I_base → max	0.50 → ~0.95	capital-task share before and after the AI ramp
K0 per capita	5.0	initial capital stock (no-automation steady state K/Y ~ 2.9)
c_income	0.85	propensity to consume out of labour income
c_profit	0.35	propensity to consume out of capital income (differential saving)
c_wealth	0.03	propensity to consume out of equity wealth
ret_sigma	0.05	cross-sectional SD of idiosyncratic wealth returns (Fagereng et al)
ret_persist	0.92	persistence of the return type
demographic_reset	0.02	turnover that pins the stationary tail
avoidance_ elasticity	0.75	base erosion w.r.t. the net-of-tax rate (Brülhart et al)
migration_semi_ elast	0.02	capital flight per pp of wealth tax (Jakobsen et al. 2024)
init_wealth_sigma	1.15	seed log-dispersion (initial Gini ~ 0.6)
gov_cost / r_debt	0.10 / 0.02	government running cost (0 in the no-government laissez-faire benchmark); interest on balances

3.7 Two-channel parameters and why they take these values

The two-channel extension replaces the single capital-task CES with a nested three-cluster CES: a routine cluster pairing routine labour with robotic capital, a cognitive cluster pairing cognitive labour with AI compute, and a top CES between the clusters. Every new parameter is set either to reproduce the single-channel baseline (so nothing in the existing results moves when the channels are switched off), to match an empirical anchor, or as a clearly-flagged illustrative policy level. The reasoning is recorded in full below so the choices are auditable rather than buried.

Parameter	Value	Why this value
e_top	1.20	Elasticity between the routine and cognitive clusters. Set above the within-cluster elasticities so robots and AI substitute more readily with each other (production can shift between routine- and cognitive-intensive output) than with their own labour, which is the nesting choice. Mild gross substitutes at the broad-task level.
e_routine, e_cog	0.60	Within-cluster capital-labour elasticities, gross complements, matching the single-channel value and the empirically defensible sub-unity range (Acemoglu-Restrepo 2022; Oberfield-Raval). Automation raises the wage on the remaining tasks in a cluster rather than collapsing it.
theta_cog	0.50	Top-level weight on the cognitive cluster. Symmetric at baseline; no prior reason to tilt, and symmetry lets the baseline match the single-channel calibration exactly.
cognitive_share	0.50	Fraction of the workforce doing cognitive work, taken as the top half by skill. Roughly the knowledge-vs-routine split of employment in an advanced economy; putting AI against the higher-skill half encodes that AI displaces white-collar work, the reverse of routine-biased robotics.
a_r_base, a_ai_base	0.50	Capital-task share in each cluster before the ramps. Set to the single-channel I_base so the no-automation economy reproduces the v3 baseline (capital share 0.255, K/Y near 3). This is the calibration anchor that makes the two-channel model collapse to the validated single-channel macro baseline; lower values starve capital of income and the economy fails to sustain itself.
a_r ramp	start 60, speed 0.05, +0.45	Robots ramp earlier and more slowly, to about 0.95. Industrial robotics has diffused for decades, so the robotic channel is the more mature, gradual one.
a_ai ramp	start 110, speed 0.10, +0.49	AI ramps later, faster, and reaches further (to about 0.99). Encodes the recent, rapid LLM wave reaching deep into cognitive tasks; the later-faster-higher profile is the empirical sequencing relative to robotics.
robot_capital_share0	0.50	Initial split of the capital stock between robots and AI compute. Symmetric start; the allocation rule then moves it endogenously.

Parameter	Value	Why this value
depreciation_r	0.05	Robotic / physical-equipment depreciation, the standard macro equipment rate, matching the single-channel model.
depreciation_ai	0.18	AI compute depreciates far faster. Frontier accelerators obsolesce in roughly two to three years and models turn over in about eighteen months, while the data-centre shell lasts fifteen to thirty years; 0.18 (about a five- to six-year effective life) is a deliberate blend of fast-obsolescing accelerators and longer-lived facilities, set well above the robotic rate. The qualitative results are robust to the exact figure.
invest_damping	0.25	Speed of the return-equalising reallocation between the two stocks: the investment share moves up to a quarter of the way toward the return-equalising target each period. Chosen for stability (no all-or-nothing flips) while still equalising net returns over several periods.
mu_frac	0.25 (illustrative)	The AI IP rent as a share of cognitive-cluster value added, the markup the model owner charges over the competitive cost of compute and labour. Anchored on the markup evidence: US markups rose from about 21% over marginal cost in 1980 to roughly 61% by 2016, concentrated in the upper-tail, IP-heavy firms (De Loecker, Eeckhout and Unger 2020). A quarter of AI value added as pure rent is a moderate level for a dominant IP owner; the markup literature is contested on magnitude (Foster-Haltiwanger-Tuttle), so this is presented as illustrative, not estimated.
s_home	1.0 vs 0.2	Fraction of AI compute located on home (or European) servers, which fixes how much of the physical AI-capital income is source-taxable. Bracketed between fully onshored and largely offshore to represent the sovereign-compute choice; the IP rent stays mobile regardless of where the servers sit.
robot_tax	0.15 (illustrative)	Source levy on robotic capital income, the Gates-style tax on the profits of labour-saving automation (Gates 2017; South Korea's 2017 incentive change; the rejected 2017 EU Parliament motion). Robots are physical and located where they operate, so the base is fully collectable.

Parameter	Value	Why this value
dst_ai	0.10 (illustrative)	A digital-services-style levy on AI revenue (value added plus rent), collected at source. In the spirit of the UK DST, a 2% levy on gross digital revenue introduced in 2020. A gross-revenue base of this kind is harder to shift offshore than profit, which is part of the rationale for such levies (HM Treasury, 2020), though it is also criticised for falling on firms regardless of profitability (ITIF, 2025). The base here is AI value added rather than gross revenue, so the rate is not directly comparable and is illustrative; the shared property that matters is reaching local value rather than shiftable profit.
tax_repat	0.30 (illustrative)	Withholding on the rent as it is repatriated as a licence fee. Double-taxation treaties commonly cap cross-border royalty and dividend withholding at low single-digit to mid-teens rates, so 0.30 sits deliberately above typical treaty ceilings to show the instrument's reach; it is illustrative, not a currently-available rate.

Three choices carry most of the weight and deserve a sentence each. The elasticity ordering (e_top above the within-cluster elasticities) is what makes the two capitals substitutes for each other but complements with labour, so automating one channel pulls investment toward it without simply collapsing the wage. The fast AI-compute depreciation is what keeps the AI capital stock from accumulating without limit and is grounded in the observed obsolescence of accelerators. And the AI rent (mu_frac) is the single addition that gives the mature economy something durable to tax: without it, competitive factor pricing competes the net return to zero and there is no permanent surplus, which is exactly why a robot tax or a corporate tax reaches so little of the AI value in steady state.

3.8 Why AI value is modelled as a mobile rent

The central modelling choice, that the durable AI surplus is a mobile IP rent rather than a return on physical capital, reflects how the providers actually operate. The scarce, hard-to-reproduce asset is the trained model and the brand, not the compute: anyone can rent broadly similar accelerators, so the return on the physical layer is competed down, while the model and its reputation command a markup. The commercial model is built around exactly this. Closed providers keep the weights behind an API and charge per unit of use, a markup over the marginal cost of inference, which is the rent in the model's sense. And it is recognised abroad: a UK or European customer of a provider such as OpenAI contracts with its Irish entity or its US parent rather than with a UK taxable presence, so the value leaves as a cross-border charge that ordinary domestic corporate tax does not reach, which is precisely the deductible licence fee the model represents. The robotic channel is treated differently, as physical capital earning a competed-down return, because robots are located where they operate and their hardware is a reproducible good, so the lasting margin there accrues to the maker abroad rather than persisting as a domestic rent.

One qualification belongs here. The rent is, at the time of writing, prospective rather than realised: the frontier labs run at a loss, with OpenAI reportedly spending well over a dollar for every dollar of revenue, and do not expect to be consistently profitable until roughly 2029 or 2030 (Anthropic targets a little earlier). The durable markup the model assumes is the mature-market structure the sector's valuations are betting on, not a description of current profit and loss, and its eventual size depends on the market staying concentrated rather than being commoditised (subsection O). The model should therefore be read as analysing the steady state the industry is investing toward, with the rent's existence and magnitude treated as a calibrated assumption about that future rather than an estimate from today.

4 Key mechanisms and why they are modelled this way

Three mechanisms do most of the work in the model, and each is a deliberate design choice with an empirical anchor rather than a convenience. Setting them out explicitly is part of justifying the approach: the wealth tax is behavioural rather than frictionless, concentration is generated rather than imposed, and the economy is closed in a way that produces a bounded, convergent transition.

4.1 1. Behavioural wealth tax and endogenous capital mobility

The wealth tax is behavioural: it erodes its own base and triggers capital flight, both calibrated to the empirical elasticities (Brülhart et al. 2022; Jakobsen et al. 2024), and the foreign ownership share responds endogenously to it. A per-agent offshore-wealth account then lets the model measure true inequality alongside the measured domestic figure. This matters because a wealth tax that assumed away avoidance and relocation would overstate its own effectiveness; building the responses in turns the central result into a tested, calibrated finding (section 6) rather than an artefact of frictionless redistribution.

4.2 2. Concentration generated from heterogeneous returns

The wealth distribution's tail is produced from heterogeneous, persistent returns to wealth, a documented empirical fact (Fagereng et al. 2020), in line with the Moll–Rachel–Restrepo (2022) return-gap mechanism. The distribution is therefore an output of the model, not an assumption baked into a kinetic kernel. This matters because a concentration result is only informative if the concentration was not hard-wired: here the same return heterogeneity that the data show is what generates the tail, so the inequality dynamics can be read as a consequence of the mechanism rather than of the modelling choice.

4.3 3. A stable, stock-flow-consistent closure

The economy is closed in the standard neoclassical way so that the automation transition is bounded and convergent. Output is supply-determined (gross complements bound potential output in capital), investment is the residual that clears the goods market (the saving-equals-investment identity made operational), and consumption follows a differential-saving rule in the Kaldor (1957) / Pasinetti (1962) tradition: a high propensity to consume out of labour income and a low one out of capital income, plus a small wealth effect on equity. Differential saving is what makes the steady state stable rather than merely calibrated: when capital becomes scarce its return spikes, and because that windfall is mostly saved it is reinvested and rebuilds the stock, so the capital-output ratio converges from any starting point. Output then grows by a large multiple over the long horizon (K/Y rising from about 2.9 to roughly 8 to 10 as the economy automates), but the path is bounded and convergent rather than explosive, and the no-automation baseline is a verified stationary equilibrium. A Sobol sensitivity analysis (section 6E) quantifies which parameters drive the results.

Scope of the claims

Consistent with the empirical debate, the model treats near-complete automation as a long-run scenario rather than a documented trend, and presents the falling labour share as one contested channel (alongside housing and market power) rather than a settled mechanism. The policy results are positive, not welfare-optimal: statements about one regime being preferable are framed in terms of measured inequality and solvency, not social welfare, since the model does not yet contain an explicit welfare criterion.

5 Method

Each experiment fixes the baseline calibration and varies one lever. Outcomes are read at the end of the 600-period horizon, averaged over the final 30 to 40 periods for distributional metrics and taken as the terminal value for stocks, then averaged across seeds with reported bands. Robustness is assessed three ways: Monte Carlo seed replication with confidence bands; a global (Sobol) sensitivity analysis decomposing the variance of the headline outcomes across the free parameters; and a formal stability analysis of the deterministic skeleton (section 6, experiment D). The no-automation baseline is a verified stationary equilibrium.

6 Experiments and results

The experiments build in a deliberate order, each one answering a question the previous one raises. The first group (A to E) re-establishes and stress-tests the single-channel distributional engine: that stock taxes compress inequality and flow taxes do not (A), that the compression is largely genuine rather than an artefact of capital fleeing offshore (B), that the concentration is microfounded rather than imposed (C), that the closure is formally stable (D), and which parameters actually drive the results (E). The second group (F to H) introduces the distinction the AI-tax debate turns on, between the foreign-owned, mobile AI IP rent and the competitive return on physical robots, and asks which tax base reaches the durable surplus and who ends up owning the capital (F, G), then makes those answers concrete for Britain (H). The third group tests the robustness and the costs of those answers: to the rent's size and to opening the economy to goods trade (I), and to the labour-market response, first as a changing labour share (J), then as measured unemployment and the funding of a safety net (L), with the AI-tax debate's two strongest objections put directly to the model in between (K) and the whole set of added parameters calibrated and sensitivity-tested (M). The final group (N to P) hardens the central recommendation against a strategic adversary and two alternatives: a foreign owner that actively shifts where the rent is booked (N), competition that erodes the rent without anyone taxing it (O), and the value of the rent as a capitalised stock rather than a flow (P). The thread throughout is a single question, which base reaches the durable surplus and who owns the automated capital, asked of progressively more demanding versions of the world.

6.1 A · The behavioural policy frontier

Question. Once the wealth tax leaks abroad and erodes its own base, on a microfounded concentration engine, under disciplined calibration, is the first-generation finding that stock taxes compress inequality more effectively than flow taxes still numerically supported under this closure?

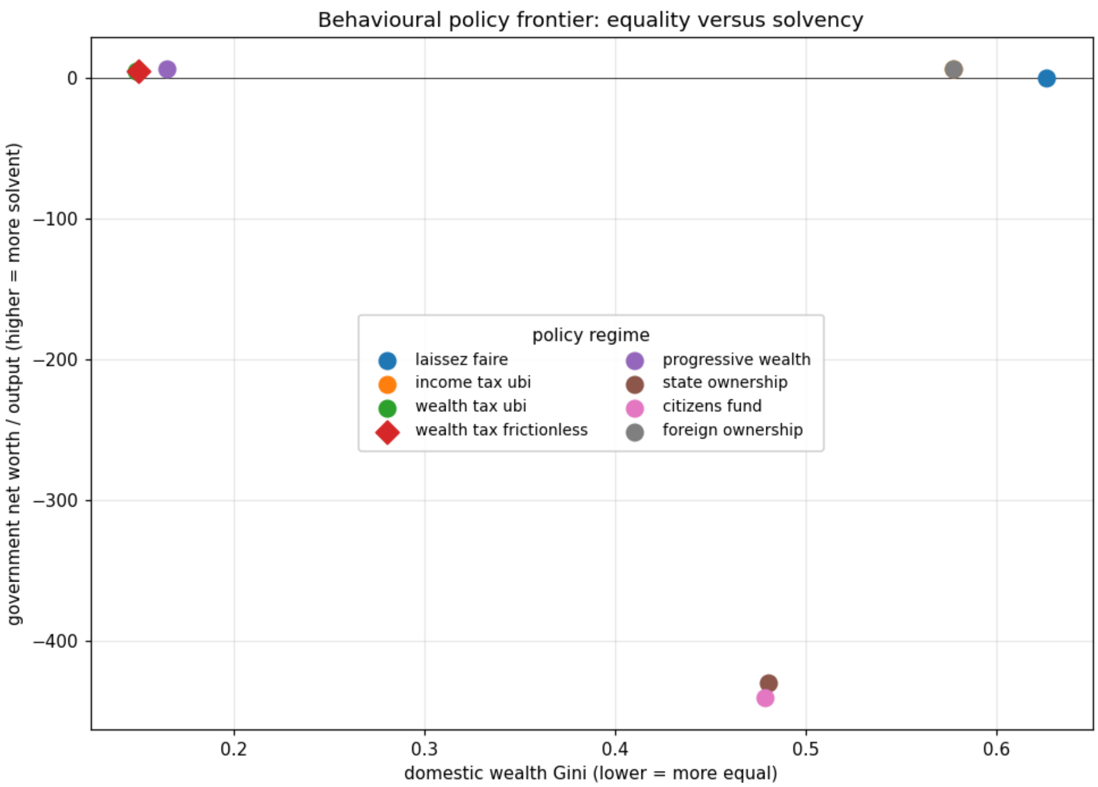

Figure 1 Behavioural policy frontier: domestic wealth Gini against government net worth relative to output. Ten-seed means. The frictionless wealth tax (diamond) sits close to the behavioural one at the baseline rate; they diverge at higher rates (Figure 2).

Result. The stock-tax regimes occupy the favourable region (low inequality, solvent): the wealth tax brings the domestic Gini to about 0.15 and the progressive variant to about 0.17, both while accumulating positive government net worth (a sovereign equity fund built from fiscal surpluses, worth roughly five to seven times annual per-capita output by the end). The flow instruments do not compress: income-tax-plus-UBI leaves the Gini at about 0.58, essentially the laissez-faire level of about 0.63. The insolvent regimes are the under-taxed welfare states: state ownership and the citizens fund pair a 15% income tax with a universal income and a 10%-of-output running cost, run persistent primary deficits, and end with deeply negative government net worth (about −420 times per-capita output) as the r-greater-than-g debt path compounds. Laissez-faire, by contrast, runs no government at all and so sits at zero net worth: it is the most unequal regime but not a fiscal one.

Interpretation. The qualitative ranking is robust within the closure: stock instruments compress the distribution, flow instruments do not. Solvency is a separate axis and turns on whether taxation is adequate to the spending commitment, not on ownership form. A redistributive state that taxes enough (the 30% income tax, or either wealth tax) not only stays solvent but gradually accumulates a majority equity stake out of its surpluses, an emergent socialisation of capital; a state that promises a universal income on a 15% income base is insolvent regardless of how much equity it nominally owns. Two caveats bound the strength of the compression result. First, part of it is mechanical: the wealth tax transfers equity in kind to the state at BOOK value, with no asset-price response, no liquidity or fire-sale discount, and no enforcement gap beyond the calibrated avoidance and migration channels, so the instrument moves ownership almost frictionlessly in a way a real wealth tax would not. Second, whether the measured equality is genuine or partly compositional (capital fled offshore but still owned domestically) depends on the rate, which experiment B quantifies. Neither overturns the ranking, but both mean the magnitudes are an upper bound on what an equivalent real-world instrument would achieve.

6.2 B · The composition effect — measured vs true inequality

Question. The wealth tax lowers the measured domestic Gini, but some of that capital has left the country rather than been redistributed. How much of the apparent equality is real, once we attribute fled capital back to its original owners?

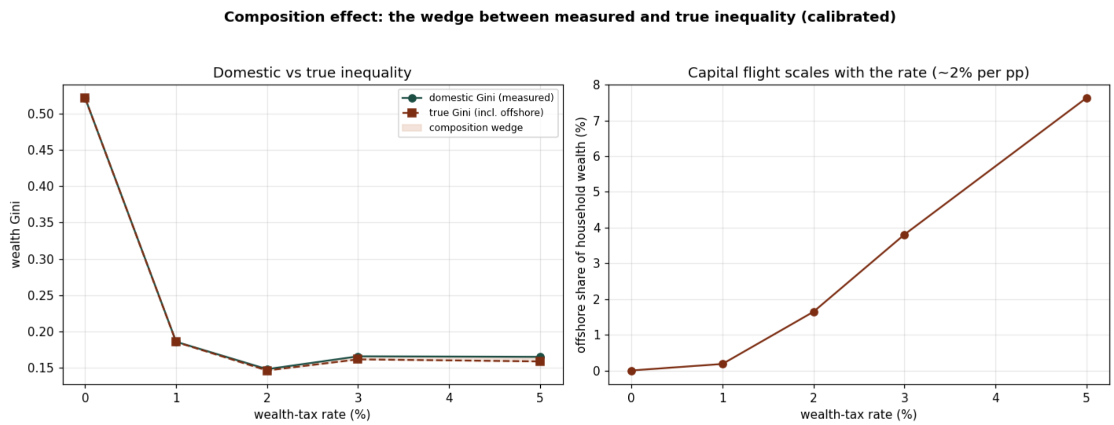

Figure 2 Left: the domestic (measured) wealth Gini against the true Gini that includes each household's offshore holdings attributed to it, across wealth-tax rates. Right: the offshore share of household wealth, which scales at about two per cent per percentage point of tax, matching the migration semi-elasticity it is calibrated to.

Result. The model tracks each household's relocated capital in a per-agent offshore account, so the true distribution can be reconstructed. The wealth tax compresses the domestic Gini sharply at the first percentage point (from about 0.63 untaxed to about 0.19 at 1%, then about 0.15 at 2%), after which further rate increases barely compress at all (about 0.17 at both 3% and 5%). The wedge between measured and true inequality stays small throughout: at a 2% tax both are about 0.15, and even at 5% the measured Gini is about 0.17 against a true Gini of about 0.16, with roughly 7.6% of household wealth offshore. The offshore share scales close to linearly with the rate (about two points per percentage point), as the calibration to the migration evidence requires.

Interpretation. The composition effect is real but modest at empirically-anchored mobility, not the dominant force an earlier, mis-calibrated version of this experiment suggested (that version let the offshore account compound without bound and implied almost the entire base fled, which is not credible). Two defensible claims follow: most of the wealth tax's compression is genuine redistribution rather than measurement artefact (the domestic-true wedge is at most a point of

Gini), and the compression is front-loaded, so a low single-digit rate captures almost all of the achievable equality while a higher rate mostly just pushes capital offshore. A careful evaluation should still report the true figure rather than the domestic one.

6.3 C · Concentration from heterogeneous returns

Question. Does the microfounded engine produce a realistic, stationary wealth distribution without the imposed kernel?

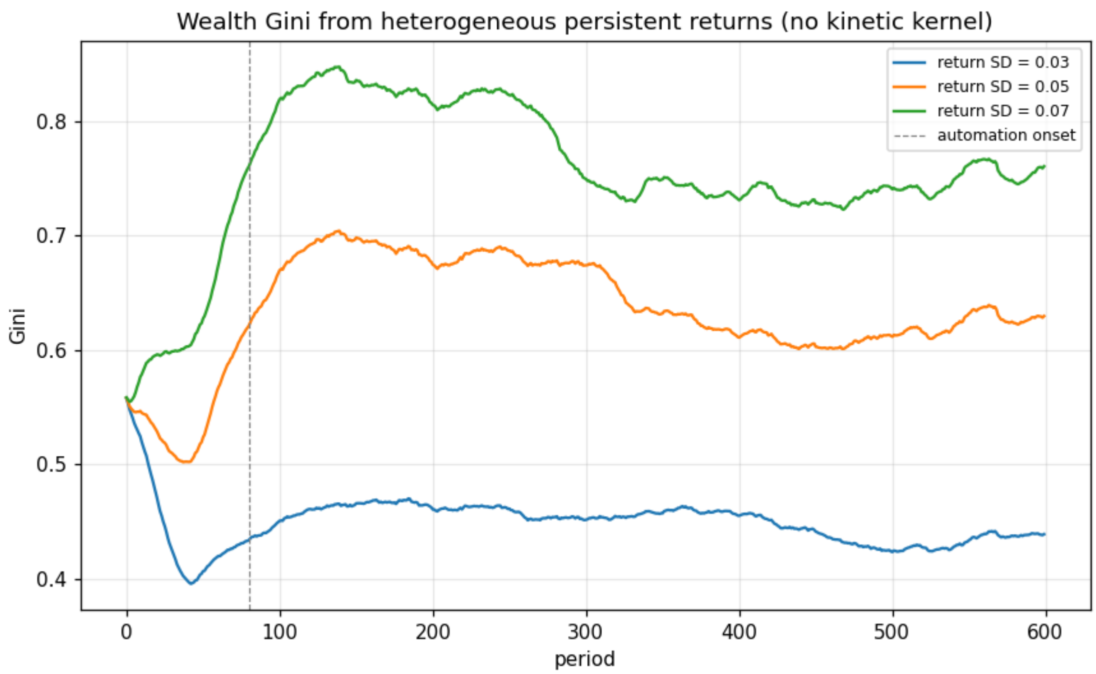

Figure 3 Wealth Gini over time for three levels of return dispersion. The tail emerges from heterogeneous persistent returns plus demographic turnover, with no kinetic kernel.

Result. The engine produces a heavy but stationary tail. Return dispersion of 0.03, 0.05 and 0.07 maps to terminal Gini of about 0.44, 0.63 and 0.75, with top-1 per cent shares of about 12, 24 and 39 per cent (and top-10 per cent shares of about 35, 53 and 68 per cent), all empirically plausible for wealth. The distribution settles rather than running away to full condensation.

Interpretation. Concentration is now an economic output driven by a calibrated primitive (the dispersion of returns to wealth), not an artefact of an imposed mean-reverting process. This removes the deepest methodological objection to the first-generation model.

6.4 D · Formal stability

Question. Is the steady state stable in a provable sense, rather than by trajectory inspection?

Result. The real economy is locally stable everywhere: the slope of the one-dimensional capital map at its fixed point lies between about 0.89 and 0.98 in every regime and at every automation level, comfortably inside the unit circle. This is the formal counterpart of the global convergence shown numerically (the capital-output ratio settles to the same value from any initial

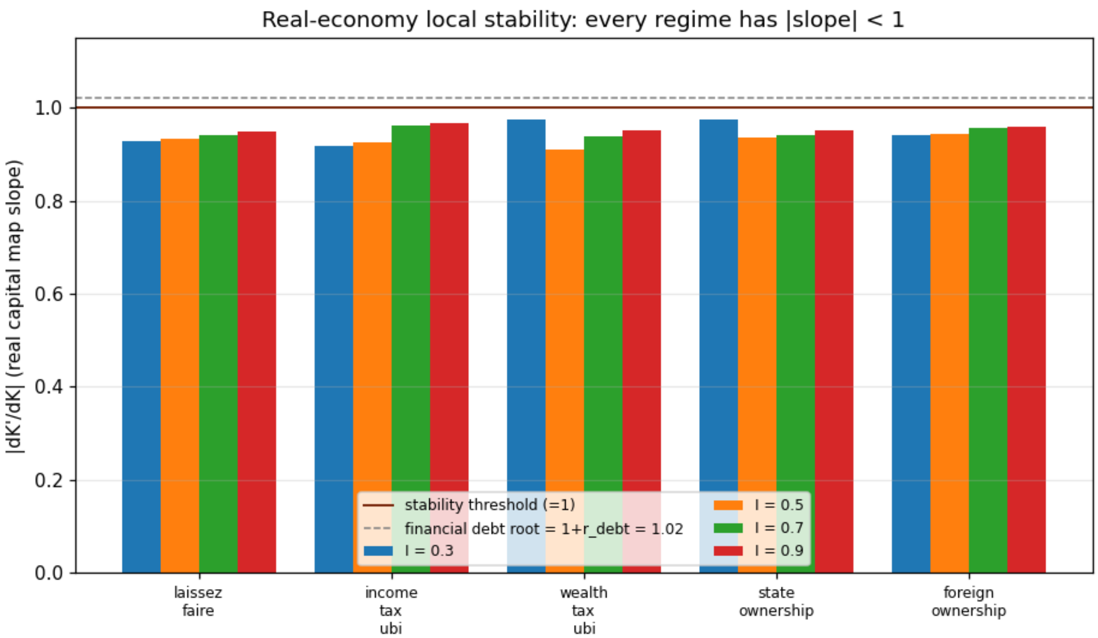

Figure 4 Local slope of the aggregate capital map, $|dK'/dK|$, at the deterministic-skeleton fixed point, across five regimes and four automation levels. Every bar is below the stability threshold of one; the dashed line marks the structural financial root 1 + r_debt = 1.02.

stock). Separately, the deposit block carries a structural root at exactly 1 + r_debt = 1.02: any bond balance compounds at the interest rate, so it collapses to one when the rate is set to zero.

Interpretation. Real stability and fiscal solvency are distinct questions. The capital stock has a stable rest point under every policy, so the real economy always converges; what differs across regimes is the financial block, where the r-greater-than-g root means that a regime running a persistent primary deficit puts public debt on an explosive path. This is the formal, root-level statement of H1: the real economy is never the source of instability, and a regime is fiscally sustainable only if it generates the surplus or the asset income to neutralise the debt root, which the adequately-taxed regimes do and the under-taxed welfare states do not.

6.5 E · Global sensitivity: which parameters drive the results

Question. Are the headline outcomes robust, or do they hinge on the least-certain parameters (the behavioural elasticities we had to assume)?

Result. At a base sample of N=64 (about 1,150 model runs) the wealth Gini is driven mainly by the dispersion of returns to wealth (total Sobol index about 0.61) and the persistence of the return type (about 0.24), with demographic turnover and the consumption-out-of-wealth propensity secondary (about 0.11 and 0.07) and the behavioural-tax elasticities small (avoidance and migration around 0.05 to 0.06 each in this sample). These behavioural indices are the least precisely estimated: at N=64 the Sobol Monte Carlo error is non-trivial (some first-order estimates come back slightly negative, the usual small-index artefact), and an independent run at the same sample size returned values nearer 0.13, so the honest statement is that the behavioural elasticities are an order of magnitude smaller than the return-dispersion driver but pinned only to within a factor of two or so

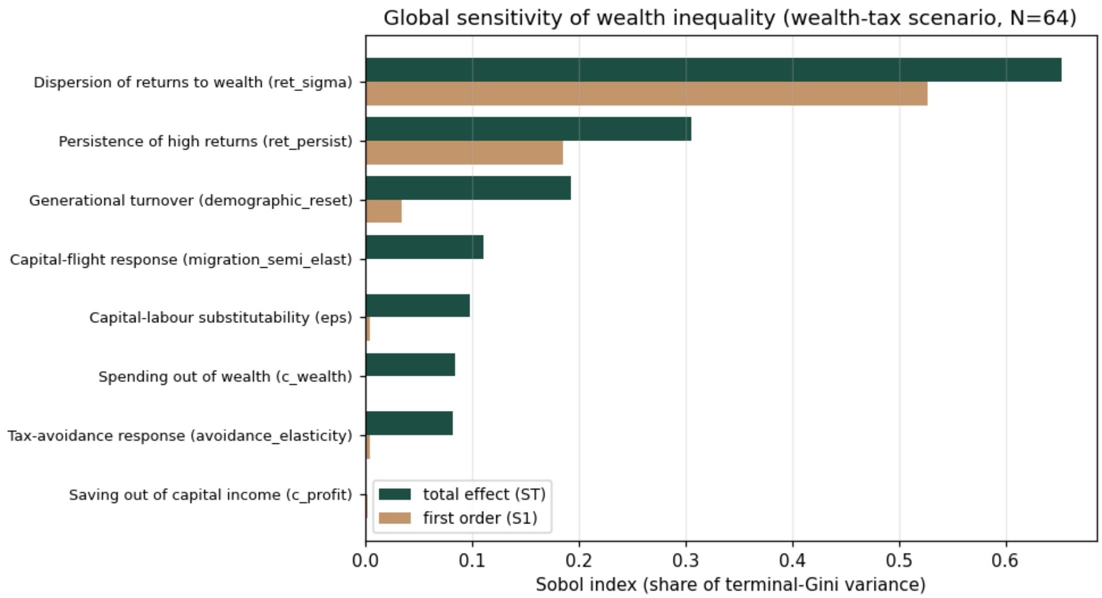

Figure 5 First-order (S1) and total-order (ST) Sobol indices decomposing the variance of the terminal wealth Gini across eight parameters (wealth-tax scenario). Higher bars are parameters that matter more.

at this sample size; a publication-grade figure would raise N. Government net worth, by contrast, is driven almost entirely by the CES elasticity of substitution (total index about 1.0, every other parameter near zero), the cleanest result in the analysis.

Interpretation. This is reassuring for the inequality results and disciplining for the fiscal ones. The distributional findings are governed by the microfounded concentration primitives (the empirical dispersion and persistence of returns to wealth) rather than by the behavioural elasticities that are hardest to pin down, so they do not rest on the least-certain numbers, even allowing for the Monte Carlo error on those elasticities. The fiscal results depend overwhelmingly on the CES elasticity, which is exactly why holding it in the empirically defensible gross-complements range (section 4) matters.

6.6 F · Source taxation of foreign-owned automation income

Question. A recurring concern for governments is that the income from highly-automated production is earned in their jurisdiction but owned, and ultimately repatriated, abroad, so a residence-based wealth or income tax never reaches it. If instead the income is taxed at SOURCE, where the output is produced, before it leaves, what happens to the ownership of the capital stock, to revenue, and to the domestic distribution?

The model levies a source tax at rate tax_repat on the foreign owners' full attributed share of after-corporate-tax profit. Crucially it falls on more than the repatriated dividend: in a growth phase profit is largely reinvested, so a dividend-only withholding tax would miss most of the income. The tax therefore has a cash leg on the part distributed abroad and an in-kind equity leg on the part reinvested in the foreign owner's name, the latter diverting that slice of new equity to the state. The proceeds are rebated to residents per capita. It is run on the foreign-ownership scenario (60% foreign at the outset).

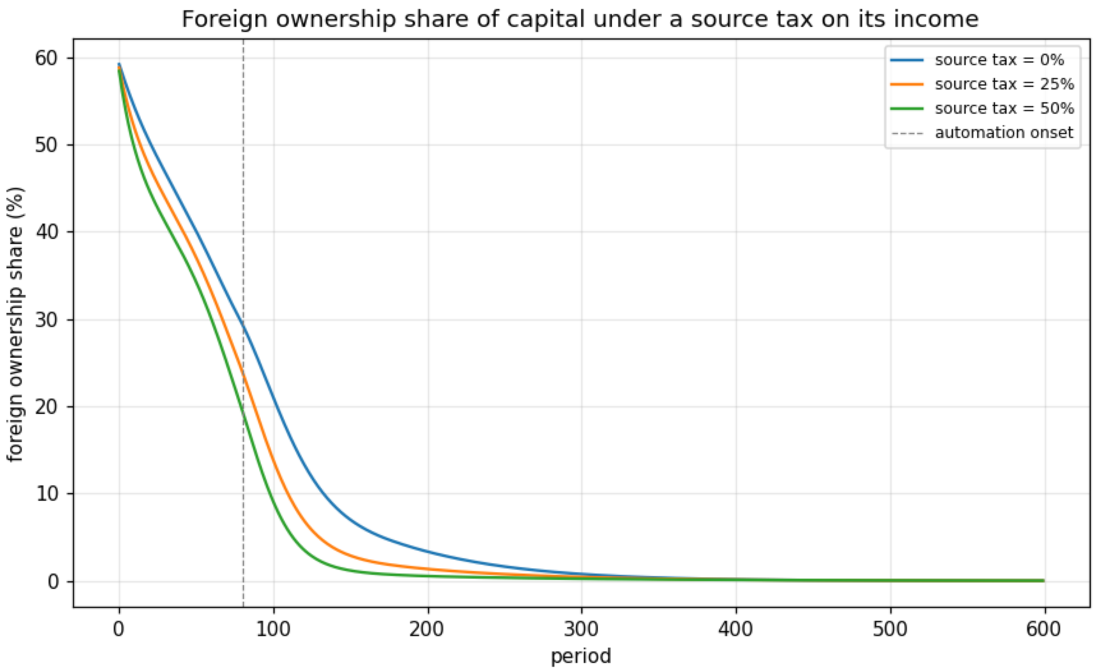

Figure 6 Foreign ownership share of the capital stock over time, under a source tax of 0%, 25% and 50% on the foreign owners' attributed automation income (proceeds rebated to residents). A higher rate brings the share down faster: the automated capital stock is domesticated sooner.

Result. The source tax raises little long-run revenue (well under one per cent of output even over the automation-transition window) and leaves the terminal wealth Gini (about 0.56) and government solvency (net worth about six times output) essentially unchanged. What it does change is the SPEED of domestication: the foreign ownership share crosses five per cent markedly sooner the higher the rate (around period 170 untaxed, 130 at a 25% rate, 113 at a 50% rate). The recycling mode (rebate to citizens versus retain as revenue) barely matters here, because the distributable foreign income is small: most of the foreign owners' return is reinvested, not paid out.

Interpretation. In this model a source tax on foreign-owned automation income is a TRANSITION instrument, not a permanent distributional lever. Domestic savers and the surplus state buy out the foreign stake regardless, so the capital stock ends up domestically owned with or without the tax; the tax mainly accelerates that, bringing the automated capital home faster and converting part of the foreign owners' reinvested earnings into a public stake along the way. Two qualifications matter for governments. First, the result is conditional on the model's dilution dynamics: it does not sustain a sticky foreign ownership share. Under structurally persistent foreign ownership (sticky FDI, which would require modelling continuous capital inflows) the source tax would be the ONLY instrument able to reach foreign owners, since the residence-based wealth and income taxes cannot, and its revenue and equality effects would then persist rather than fade. Second, the in-kind equity leg is what gives the tax its bite: a conventional dividend withholding tax, taxing only repatriated profit, would capture almost nothing while earnings are being reinvested, which is precisely the automation-investment phase governments would most want to tax. The policy reading is that taxing automation income at source is valuable mainly for the speed and the politics of domestication, and only becomes a first-order revenue and equality instrument if foreign ownership is genuinely persistent and the levy reaches reinvested, not just repatriated, earnings.

6.7 A domestic owner first: the simpler benchmark (the US case)

Before the foreign owner that occupies the rest of this section, take the simpler case the United States is in: the AI is owned at home. The same two-channel economy applies, but the AI IP rent no longer leaves as a licence fee, it accrues to resident owners as capital income (the model's foreign-owned-IP share set to zero). Nothing leaves the country, so the worry is not lost ownership but how the rent is distributed at home.

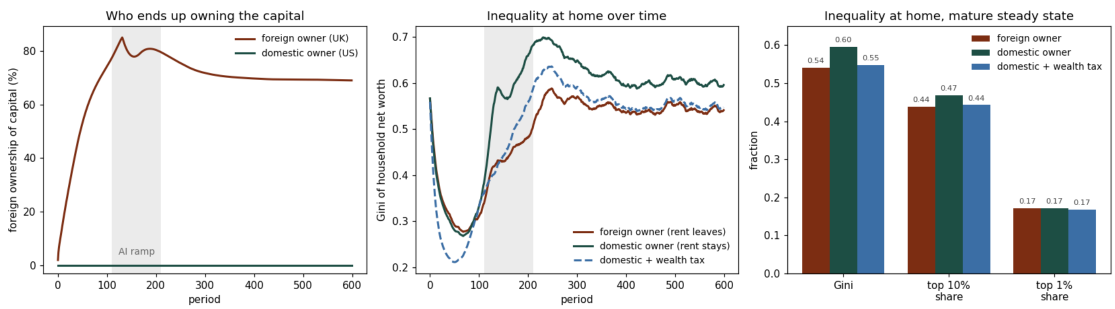

Figure 25 The two owner regimes, the same economy, with only the owner's domicile changed. Left: foreign ownership of the domestic capital stock. With a foreign owner (the UK case) the reinvested rent carries ownership toward seventy per cent; with a domestic owner (the US case) it stays flat, because the rent never leaves to be recycled into foreign-held equity. Centre: inequality of household net worth over time. The domestic owner keeps the rent at home but concentrates it among capital owners, so home inequality runs higher than under the foreign owner, where the surplus simply leaves the country; an illustrative five per cent wealth tax, which can reach the home-held rent because the owner is resident, compresses it back to about the foreign-owner level. Right: three measures at the mature steady state. The owner-domicile effect shows up in the Gini (about 0.54 versus 0.60) and the top-decile share (about 44 versus 47 per cent), but not in the top-percentile share (about 17 per cent in every case), because the rent accrues pro-rata to equity and so concentrates among the broad owning class rather than the very apex.

Interpretation. Two things follow, and they are the mirror image of the foreign-owner results below. First, ownership does not drift abroad, so the national-ownership problem that dominates the rest of this section does not arise. Second, the rent concentrates among the domestic households that own the AI, so the problem becomes domestic inequality, and the instruments that bite are the ordinary domestic ones: a wealth tax, capital-gains and dividend taxation, and competition policy. The concentration is broad rather than apex: it raises the Gini and the top-decile share but leaves the top-percentile share essentially unchanged (Figure 25, right), because the rent accrues in proportion to equity holdings, so it enriches the owning class as a whole rather than only the very richest. Crucially, a wealth tax *does* reach this rent, because the owner is resident and the IP is onshore, which is exactly the lever that fails against a foreign owner (subsection N and section 8). The cost is the one specific to the provider's home: this is the regime in which the territoriality objection of section 1 has force, since taxing a domestic champion can push its activity to a lighter-touch jurisdiction. The policy menu here is therefore the familiar one of domestic progressive capital taxation, traded off against competitiveness.

This benchmark is deliberately compact, fixing the simpler case so the contrast is clear. The harder and more novel case, and the one a country like the United Kingdom actually faces, is a *foreign* owner, where the rent crosses a border and the domestic levers above no longer reach it. Experiments G to P take that case, and it is the paper's main contribution.

6.8 G · AI versus robotic automation, taxed with different instruments

Question. The single-channel model treats all automation alike. In reality the two dominant forms sit in very different places for tax. Robotic automation is physical capital located where it operates, so its income is taxable at source. AI automation is split: the compute (servers, data centres) is locatable physical capital, but the value sits in mobile intellectual property that earns a monopoly rent and is recognised wherever the parent chooses. If a foreign multinational runs the AI layer, charges a deductible licence fee, and books the rent abroad, which instruments can a host state actually use, and what do they reach?

The production block is replaced by a nested three-cluster CES (production_v4): a routine cluster pairing routine labour with robotic capital, a cognitive cluster pairing cognitive labour with AI compute, and a top CES between them with a higher elasticity, so robots and AI substitute more with each other than with their own labour. Robots displace routine workers, AI displaces the higher-skilled cognitive workers. The two capital stocks accumulate from the same investment pool, allocated by a damped move toward equalising their net returns, with AI compute depreciating faster. A markup peels a share of cognitive value added as an IP rent that leaves to the foreign owner as a deductible licence fee, so it never enters the corporate base. Four instruments are compared on this structure: a source tax on robotic income, a withholding on the rent, a digital-services levy on AI revenue, and the home-located share of compute.

scenario	AI rent leaving (% output)	capital-sector revenue (% output)	govt net worth (× output)	wealth Gini	foreign ownership of capital
foreign AI rent, untaxed	13.1	0.0	0.9	0.544	0.69
+ robot tax (15%)	13.1	1.5	1.5	0.540	0.64
+ AI digital-services levy (10%)	13.1	3.7	2.8	0.535	0.51
+ rent withholding (30%, rebated)	13.2	4.0	0.9	0.506	0.40
compute offshore (s_home 0.2)	13.1	0.0	0.9	0.544	0.69
full domestic toolkit	13.1	5.2	3.2	0.531	0.48

Steady-state averages (periods 250 to 600, five seeds). Capital-sector revenue aggregates corporate tax, the robot tax, the digital levy, and the rent withholding. Foreign ownership is the rest-of-world equity share of the capital stock at the horizon.

Result. The AI layer generates a permanent rent of about thirteen per cent of output that, untaxed, leaves the country every period. Corporate tax reaches almost none of it, because the rent is a deductible licence fee and the competitive capital profit is competed to zero in the mature economy. The robot tax cleanly collects on the physical robotic income (about 1.5 per cent of output at a fifteen per cent rate) but cannot touch the rent. The digital-services levy is the effective instrument on the rent, collecting 3.7 per cent of output and lifting government net worth from 0.9 to 2.8 times output. The rebated withholding collects a similar amount but, because it is handed straight back to citizens, raises no net public saving while delivering the lowest inequality. The compute-location share moves the transition but not the steady state, because the durable surplus there is the rent, which the levy reaches wherever the servers sit. The deepest result is the

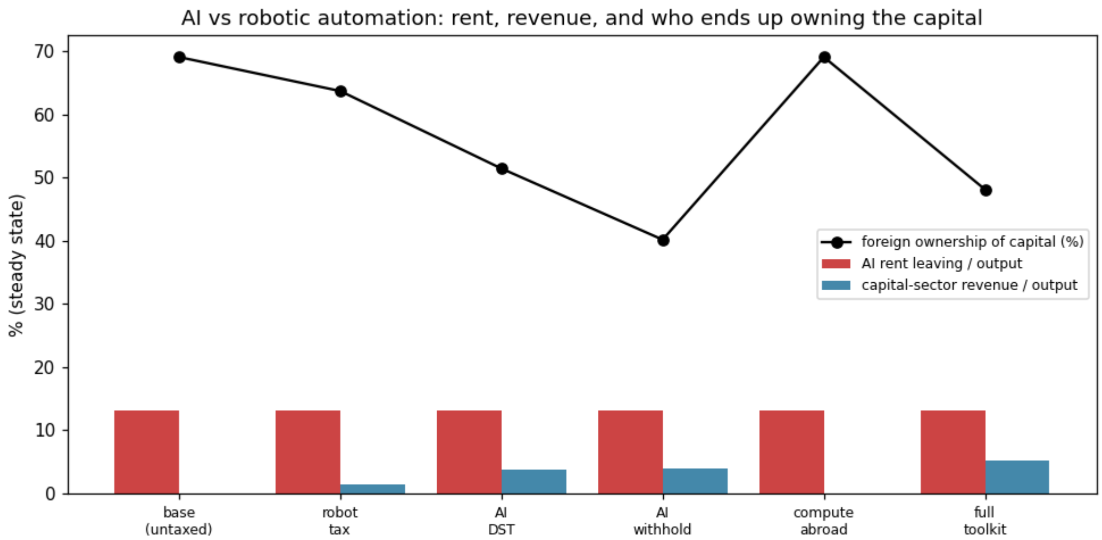

Figure 7 Under a foreign-owned AI rent, what each instrument reaches in the mature economy: the rent leaving the country (red), the capital-sector revenue collected (blue), and the foreign share of the capital stock (black line). The digital levy is the effective revenue instrument; the rebated withholding is the most equalising and most curbs foreign ownership; corporate tax and the compute-location share reach almost nothing in steady state.

foreign-ownership column: left untaxed, the AI owner recycles its rent into buying domestic equity and ends up holding about seventy per cent of the capital stock; taxing the rent slows that buy-up to roughly half (levy) or forty per cent (rebated withholding), because it leaves less rent to reinvest.

Interpretation. Splitting the two channels changes the policy conclusion. A robot tax is administrable and lands squarely, but it taxes the wrong thing: the physical, competitive, low-rent part of automation, while the durable surplus is the mobile AI rent it cannot reach. Only a levy on AI revenue, or a withholding that reaches the licence fee itself, touches that rent, and the corporate tax and any compute-onshoring incentive are second-order in the mature economy (they matter during the transition profit window, not after). The most consequential finding is about ownership rather than revenue: a persistently extracted, reinvested AI rent does not merely drain a flow, it lets the foreign owner accumulate the domestic capital stock, so the instruments' deeper value is curbing that drift. Two caveats bound this, and subsection I takes both up directly. The ownership result is conditional on the closure: with no goods-trade channel, the rest of the world can only spend its rent receipts on host assets, so the model shows the fully-reinvested pole; adding a trade balance (Figure 15) lets the rent be repatriated as real resources, which turns the single ownership number into a range and drains the flow without necessarily transferring ownership. And the rent itself is a calibrated markup rather than an estimated one, so the magnitudes illustrate the mechanism rather than forecast a number, though the ranking of instruments is robust to the markup (Figure 13).

6.9 H · Worked examples: what each policy means for Britain, its people, and the foreign owner

To make the table concrete, read the host economy as the United Kingdom. The robots in the routine cluster are supplied and largely owned by a foreign humanoid-robotics maker, a Tesla

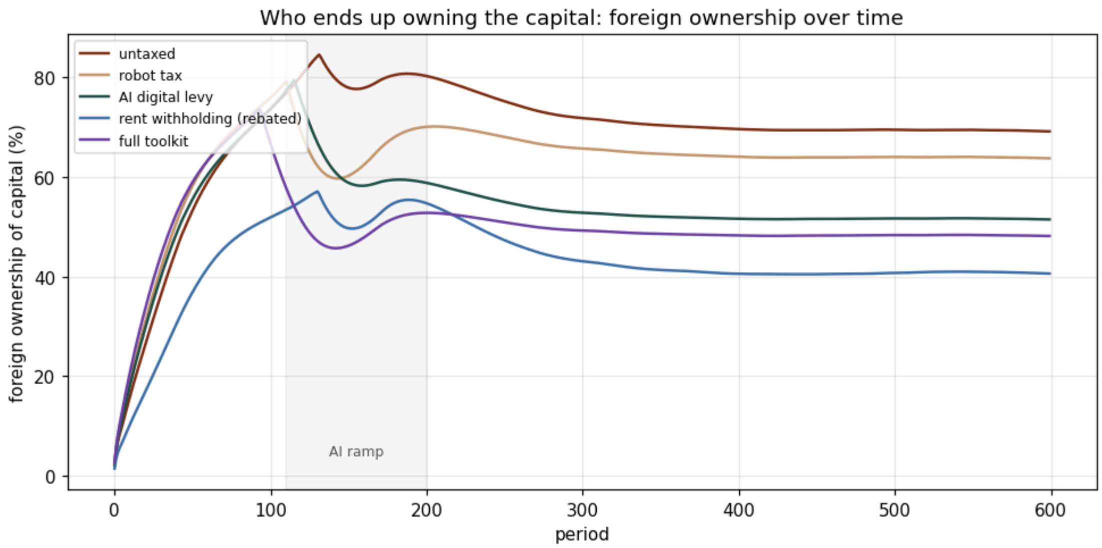

Figure 8 Who ends up owning the capital. Foreign ownership of the domestic capital stock over time under five policies, with the AI ramp shaded. Left untaxed, the foreign owner recycles its rent into domestic equity and climbs toward eighty per cent of the stock during the transition, settling near seventy per cent. Each instrument that reaches the rent leaves less to reinvest and so holds ownership lower in the long run: the digital levy to about half, the full toolkit to just under half, and the rebated withholding furthest, to about forty per cent. The robot tax, which misses the rent, barely shifts the path.

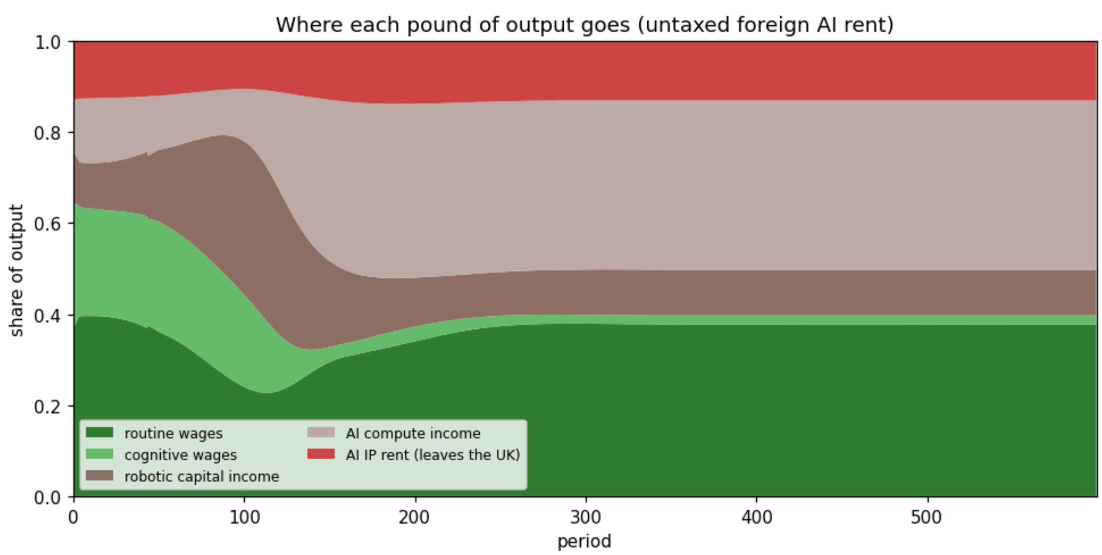

Figure 9 Where each pound of output goes, over time, under the untaxed baseline. As automation proceeds the labour share falls (the cognitive wage share nearly vanishes as AI takes the cognitive cluster, while routine wages hold up better), and the gains accrue to AI compute income and to the AI IP rent. The rent (red, about thirteen per cent of output in the mature economy) is the band that leaves the country every period; it is the surplus the digital levy and the withholding are designed to reach, and the part that ordinary capital and corporate taxes miss.

(Optimus) or a 1X (NEO); the AI in the cognitive cluster is supplied by a foreign frontier-model company, an OpenAI or an Anthropic, whose model weights and brand are the intellectual property. The foreign owner is resident in the United States or China. UK workers operate alongside both, UK consumers and firms buy the automated output, and the UK Treasury tries to tax the value. Each scenario below is a specific policy (a parameter setting) and what it does to four parties: the UK economy, the UK population, the UK state, and the foreign owner.

Baseline: a foreign-owned AI rent, left untaxed (mu_frac 0.25, no robot tax, no DST, compute at home). The AI company prices its service well above the cost of the compute and the cognitive labour it uses, and books the difference, about thirteen per cent of national output every period, as an IP licence fee paid from the UK operation to the parent abroad. *UK economy:* output still rises through automation, but a permanent slice leaves as a deductible fee, so measured domestic income is lower than the production would suggest. *UK population:* cognitive workers (the higher-skilled half) see AI take a growing share of their cluster; measured domestic inequality actually falls, because the person capturing the rent sits in San Francisco or Shenzhen, not in the UK top tail. *UK state:* the corporate tax base is nearly empty, because the rent is a deductible cost and the competitive return is competed away; the government's sovereign fund slips to under one times output. *Foreign owner:* collects the rent and recycles it into buying UK equity, ending up owning roughly seventy per cent of the domestic capital stock. The quiet result is loss of ownership, not just a tax gap.

Policy 1: a robot tax (robot_tax 0.15). A Gates-style levy on the income of the physical robots operating in Britain. *UK economy and state:* it lands cleanly and raises about 1.5 per cent of output, because robots cannot be moved out of the jurisdiction they work in; the fund recovers toward 1.5 times output. *UK population:* modest extra revenue to fund transfers. *Foreign owner:* the robot maker pays, but the AI rent is untouched, so the owner still accumulates most of the capital stock. The lesson: a robot tax is administrable and fair on the physical channel, but it taxes the part of automation that carries little rent and misses the part that carries most.

Policy 2: a digital-services levy on AI (dst_ai 0.10). A tax on AI revenue collected in Britain, in the spirit of the UK's existing 2% digital-services tax, which was designed to tax revenue precisely because revenue is far harder to move offshore than profit. *UK state:* this is the effective instrument, collecting about 3.7 per cent of output and lifting the fund back toward 2.8 times output, because it reaches the rent the corporate tax cannot. *UK economy and population:* the rent leaking abroad shrinks and inequality edges down. *Foreign owner:* with less rent to recycle, its accumulation of UK capital slows from seventy to about fifty per cent. This is the lever that actually bites on the AI side, and it is the one that has angered the United States in the real DST disputes.

Policy 3: a withholding on the rent, rebated to citizens (tax_repat 0.30, rebated). Britain intercepts a share of the licence fee as it leaves and hands it straight back to residents per head. *UK population:* the most equalising option, because it is a direct transfer to citizens; the wealth Gini falls furthest. *UK state:* raises no net public saving, since the proceeds are rebated rather than banked, so the fund does not recover. *Foreign owner:* the buy-up of UK capital is curbed the most, to about forty per cent, because the rebate strips the most rent out of the reinvestment loop. A real version would run into treaty caps on withholding rates, so the 30% here is illustrative of the reach rather than a currently-available rate.

Policy 4: where the compute sits (s_home, 1.0 versus 0.2). Whether the AI is served from data centres in Britain or Europe, or from the United States. *Steady state:* in the mature economy this barely moves revenue, because the durable surplus is the mobile IP rent (reached by the DST), not the competitive return on the servers. *Transition:* it matters more while capital returns are briefly high, and it is the lever a sovereign-compute policy pulls. The reading for Britain is that onshoring compute is worth doing for resilience and for the transition tax base, but it is not a substitute for a levy that reaches the rent itself.

Policy 5: the full domestic toolkit (onshored compute, robot tax and DST together). *UK state:* raises the most, about 5.2 per cent of output, and restores the fund to over three times output. *Foreign owner:* ownership drift is held to about half the capital stock. *Reading:* the instruments are complements, not substitutes: the robot tax catches the physical channel, the DST catches the AI rent, and onshored compute widens the transition base, so a state that wants both revenue and to keep ownership at home uses all three rather than betting on one.

6.10 When the rent rises with automation: pricing power and output capture

Question. The baseline holds the rent at a fixed share of cognitive value added, which keeps it at a roughly constant eighth of output throughout. That is deliberately conservative, and it embeds an assumption worth testing: that automation changes the composition inside the cognitive cluster (AI compute income replacing cognitive wages) without changing the owner's cut or the cluster's share of the whole. There are two reasons the rent share could instead climb as automation deepens. First, pricing power: as the model becomes more essential and the market more concentrated, the owner can charge a higher markup, not a fixed one. Second, output capture: AI need not stay a priced input inside an otherwise-human firm. AI-native producers can do the whole job and displace the firms that used human workers, so the AI cluster takes a growing share of total output rather than a fixed slice of one cluster. The early evidence in software, where AI is capturing a rising share of development work rather than merely assisting it, is of the second kind. Two off-default levers represent these: a markup that rises with the degree of AI automation, and a top-level weight on the cognitive cluster that rises with it.

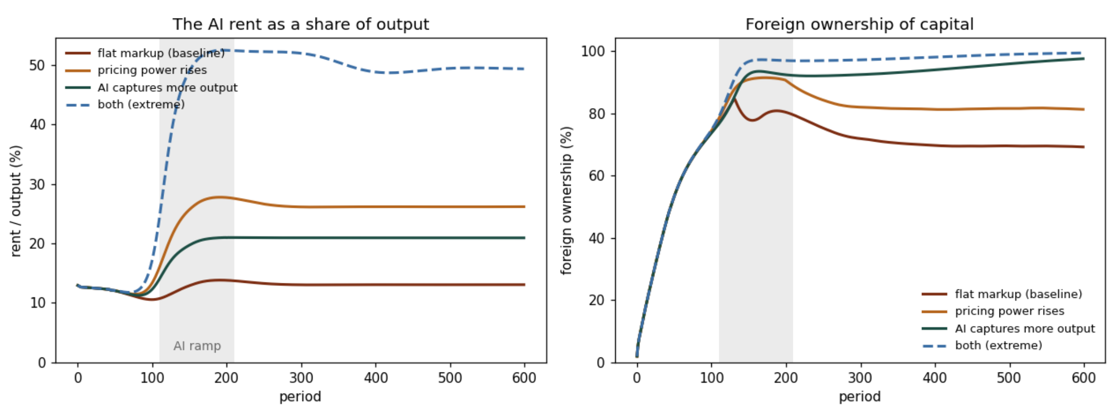

Figure 26 The rent need not stay a fixed share. Left: the AI rent as a share of output. Held flat (the baseline) it settles near an eighth; with pricing power rising over the automation ramp it roughly doubles to about a quarter; with the AI cluster capturing a growing share of total output it reaches about a fifth; with both it approaches half of output. Right: the consequence for ownership. The flat-rent case plateaus near seventy per cent foreign ownership, but a rising rent accelerates the drift toward the whole capital stock, eighty-two per cent under pricing power and ninety-five to ninety-nine per cent once the AI cluster captures output or both channels operate.

Interpretation. The flat-markup baseline used everywhere else in the paper is therefore the conservative floor, not the central case. If the rent rises with automation, which is the more likely direction and is arguably already visible, every result in the paper sharpens rather than softens: the surplus at stake is larger, the foreign owner accumulates the domestic capital stock faster and more completely, and the cost of delay and the case for reaching the rent early both grow. A rent-importer that finds the flat-rent ownership drift to seventy per cent uncomfortable should note that the realistic risk is closer to the whole stock.

One honest structural limit bounds how literally to read the output-capture channel. The model still has a single representative firm with a markup, so the capture lever raises the cognitive cluster's share of output in reduced form rather than mechanising the displacement. It approximates the economics of AI-native production eating into the human cluster, but it does not model competition between distinct AI-native and human-incumbent firms, in which an AI producer would capture the entire output of the firms it displaces rather than a markup on a cluster. That heterogeneous-firm treatment is the proper way to model the distinction between AI used as an input and AI used as the business, and it is the natural next structural extension; this subsection bounds the consequence of moving in that direction without yet building the mechanism. The owner's domicile is orthogonal to all of this: the rising-rent levers change how large the surplus is, while the foreign-owned-IP share (the domestic-owner benchmark) changes where it goes, and a rising rent makes the domicile question more consequential, not less.

6.11 Robots embody IP too: the rent is on the model, not the machine

Question. The baseline puts the entire rent on the cognitive cluster and treats the robotic channel as purely competitive: reproducible hardware earning a normal return that ordinary instruments, and a hardware robot tax, can reach. That is the cleanest case for the robot-tax debate, but it is not the only one. A robot is hardware plus the IP that runs it, the control policy or foundation model and the proprietary design, and as robots become more capable that IP is the scarce, foreign-held part in the same way the AI model is, while the metal is a commodity anyone can buy. An off-default lever (robot_ip) puts a foreign-held markup on the robotic cluster, peeled from its value added exactly as the AI markup is peeled from the cognitive cluster and routed through the same licence-fee, withholding and digital-levy machinery. The hardware robot tax continues to fall only on the competitive remainder.

Interpretation. With a robot markup equal to the AI one, the total foreign-held rent rises from about an eighth of output to about a quarter, and the ownership drift accelerates from roughly seventy per cent of the capital stock toward almost the whole of it. The instrument result is the point. The hardware robot tax barely touches the drift, because it lands on the competitive hardware return while the rent sits on the IP, but the source levy reaches the robot rent exactly as it reaches the AI rent, because both are recognised as the same cross-border licence fee. The ranking is therefore unchanged, and the robot tax's role is reinforced as a thin complement on the physical channel, not the instrument that reaches the surplus. The deeper reading is that the economically meaningful distinction is not robots versus AI but competitive hardware versus rent-bearing IP. In the limit the two channels converge into a single thing, a model that does a growing share of total labour, physical and cognitive alike, with the durable rent sitting on the model wherever it is embodied. Confining the rent to the cognitive cluster, as the rest of the paper does, therefore understates the surplus rather than overstating it.

The same structural limit as the rising-rent levers applies. The robot rent is a markup on a single representative firm's robotic cluster, foreign-held in the same proportion as the AI rent,

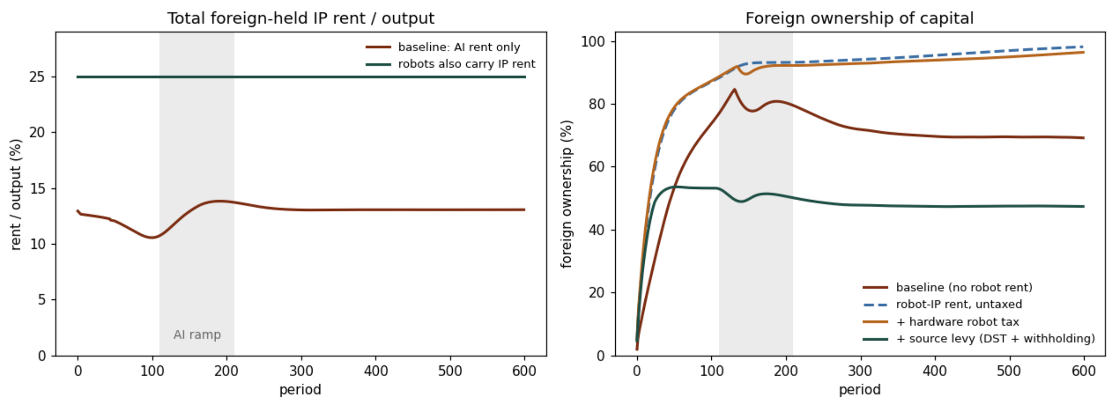

Figure 27 Robots carry IP too. Left: the total foreign-held IP rent as a share of output. With the rent confined to the AI cluster (the baseline) it settles near an eighth; giving robots an IP markup of the same size as the AI one roughly doubles it, to about a quarter. Right: foreign ownership of the capital stock under that robot-IP rent. Untaxed it drifts toward the whole stock, and a hardware robot tax barely changes that path (the two lines almost coincide), because it falls on the competitive hardware return and the rent has already been peeled away from it. The same source levy that reaches the AI rent (a digital levy plus a withholding) reaches the embodied robot rent and holds ownership well below even the no-robot-rent baseline.

rather than the outcome of competition between rent-bearing and commodity robot producers; and the incentive cost of taxing it is kept on the AI side, so the figure is conservative about the efficiency cost of reaching the larger rent. The lever is off by default, so every other result in the paper is the special case in which robots earn no rent.

6.12 I · Robustness and two open-economy extensions

The two-channel interpretation (subsection G) carried two explicit caveats: the conclusions might depend on the size assumed for the rent, and the ownership result was the fully-reinvested pole of a closed economy. This subsection addresses both, first by checking robustness to the rent size and to simulation noise, then by relaxing the closure and adding the missing incentive cost. Each extension is a single parameter that is off by default, so every result above is the special case in which it is zero.

Robustness. The rent markup is calibrated, not estimated, so the first question is whether the ranking of instruments survives a different rent. Sweeping the markup from a tenth to two fifths of cognitive value added moves the levels (a larger rent means more foreign ownership and more revenue to collect) but leaves the ordering intact at every point: the digital levy and the full toolkit always reach the rent, the robot tax always misses most of it, and the untaxed baseline always lets ownership drift furthest. The conclusion is about which lever reaches the rent, and that does not depend on how large the rent is.

Simulation noise is the second robustness question. Re-running the five headline policies over sixteen seeds, the capital-sector revenue is effectively deterministic (confidence intervals under a tenth of a percentage point) and the foreign-ownership share varies by at most about two tenths of a percentage point across seeds. The headline numbers are properties of the structure, not artefacts of a particular draw.

Open economy. The ownership result was the closed pole: with no goods-trade channel the

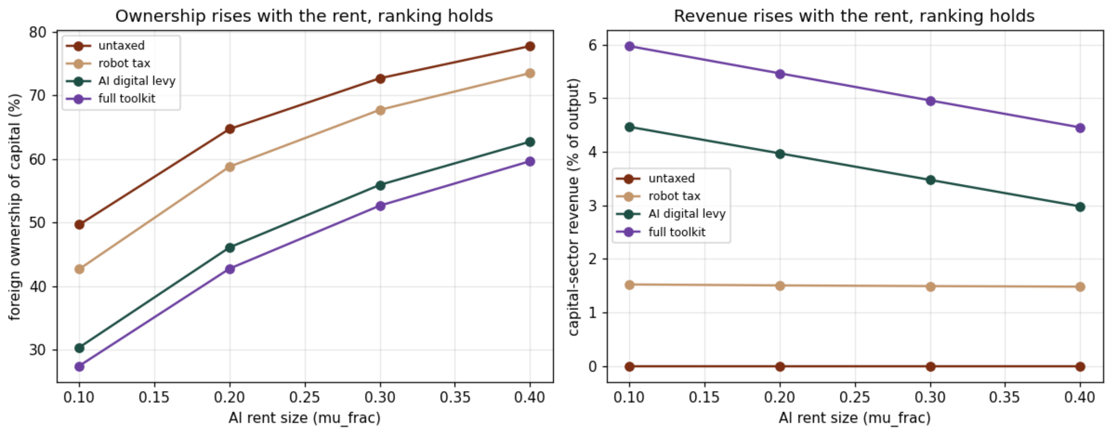

Figure 13 Robustness to the rent size. Foreign ownership of the capital stock (left) and capital-sector revenue (right) as the AI markup mu_frac varies from 0.10 to 0.40, for four policies. The levels rise with the rent but the ordering of the instruments is preserved throughout, so the qualitative conclusions do not hinge on the calibrated markup.

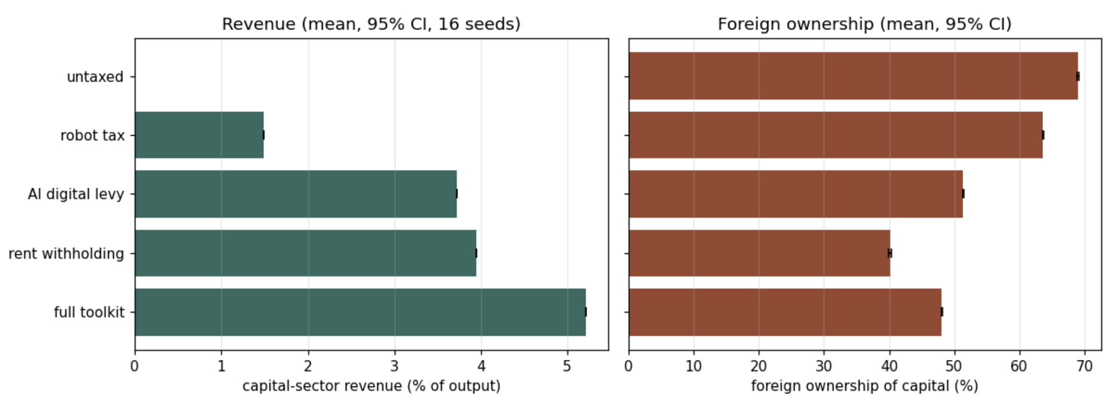

Figure 14 Seed dispersion. Capital-sector revenue (left) and foreign ownership (right) as means with 95 per cent intervals over sixteen seeds. The intervals are within the marker width, confirming the headline comparisons are not seed-sensitive.

foreign owner can only spend its rent on host assets, so it accumulates the capital stock. Adding a current-account leak, the share of the after-tax rent the owner repatriates as real goods rather than reinvesting, turns the single number into a range. At one extreme (no repatriation) the owner reaches about seventy per cent of the stock, as before; at the other (full repatriation) it reaches none, because the rent leaves as imports and buys no equity. The realistic case lies between, and the same channel carries a real cost: repatriating the rent as goods means less is invested at home, so the steady-state capital stock and output fall to about three-quarters of the closed economy. The drain and the ownership transfer are therefore distinct harms, and trade policy moves the economy along this line rather than removing the problem.

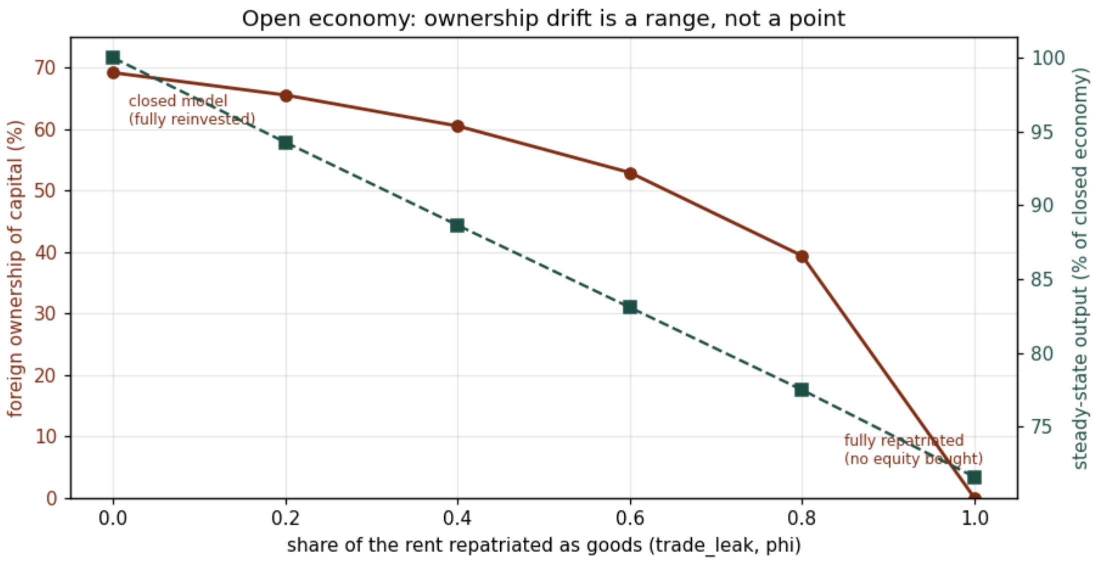

Figure 15 The ownership result as a range. As the repatriated share of the rent rises from zero (the closed, fully-reinvested model) to one (the rent taken entirely as goods), foreign ownership of the capital stock falls from about seventy per cent to zero, while steady-state output falls to roughly seventy per cent of the closed economy. Repatriation drains the flow and shrinks domestic capital instead of transferring ownership: the two are separate costs, and the truth lies between the poles.

Incentive cost. The instruments above reached the rent without the owner reacting, which made taxing it look free. Allowing the AI owner to deploy less capability as its net-of-tax rent falls gives the levy an efficiency cost. Under the digital levy and a repatriation withholding, a modest supply response lowers steady-state output to about four-fifths of the untaxed level, while the revenue collected as a share of output barely moves. The reading is not that the levy fails to raise money, it still does, but that it shrinks the pie it taxes, so the case for reaching the rent must be set against an output cost rather than treated as a free transfer. The size of that cost is illustrative: the model's accumulation channel amplifies any persistent output loss, so the per-period deadweight is capped to keep the long-run magnitude credible, and pinning it down empirically is left to the next phase.

Calibration. The two extensions above are now anchored to data rather than set arbitrarily, with the full sweeps (Figures 15 and 16) bracketing the range. The repatriation share is set to a half: balance-of-payments evidence puts reinvested earnings at roughly forty to sixty per cent of foreign direct-investment income for mature host economies (higher for young investments, by the FDI life-cycle), so the complementary repatriated share sits near a half; and because the AI rent

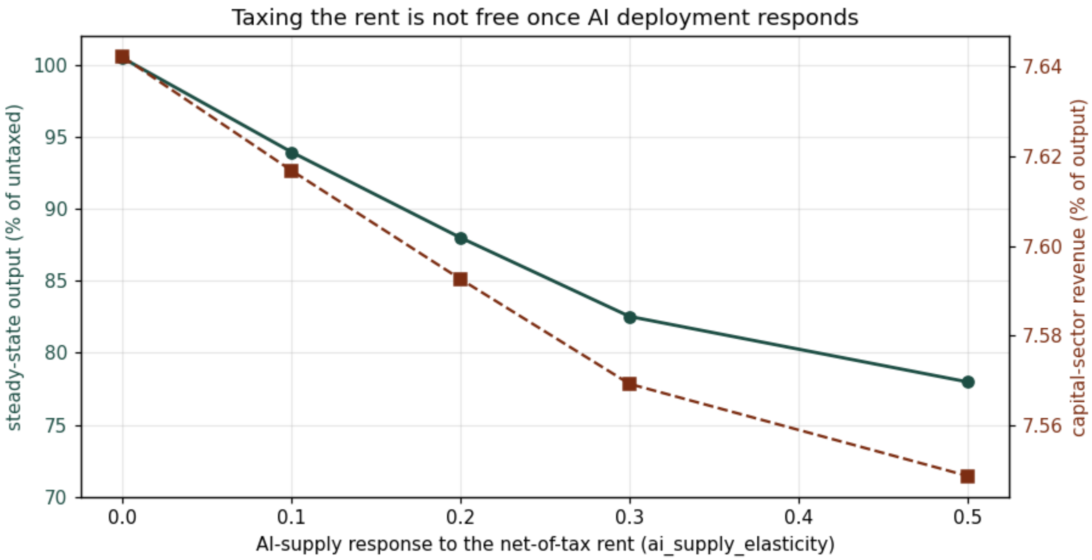

Figure 16 The cost of taxing the rent once AI deployment responds. As the AI-supply elasticity to the net-of-tax rent rises, steady-state output falls to about seventy-eight per cent of the untaxed level (teal), while capital-sector revenue as a share of output is almost unchanged (sienna). Taxing the rent is no longer free: it leaves the take broadly intact but lowers the level of output, a trade-off the policy choice must weigh.

is a licence fee, which is by nature a distributed flow rather than reinvested earnings, a half is if anything conservative. The AI-supply elasticity is set to 0.5, the low end of the empirical user-cost elasticity of investment, which the literature places at about −0.5 to −1.0 (Hassett and Hubbard 2002; firm-level estimates cluster −0.25 to −0.75). These are central values for illustration, not estimates for this market, which has no tax-variation history to estimate from; the point is that the mechanism's direction is now disciplined by plausible magnitudes.

6.13 J · Reinstatement and the labour share

Labour market. The model so far lets automation push the labour share down with no countervailing force, so that fall is partly mechanical. Adding a reinstatement margin (new labour-intensive tasks created in the wake of automation, in the sense of Acemoglu and Restrepo) lets the share recover. New tasks are taken to be as productive as the automated ones, so reinstatement is output-neutral at the technology level and moves income from capital to labour rather than destroying it; the only general-equilibrium effect on output is indirect, through a lower aggregate saving rate as income shifts to higher-spending households. The result is that the labour-share path is conditional, not mechanical: it dips through the AI ramp and then settles at a level set by how strongly new tasks offset automation, from about forty per cent with no reinstatement to about sixty per cent in the balanced case. The trade-off is that a more labour-heavy economy saves and accumulates less, so steady-state output eases as reinstatement strengthens.

6.14 K · Testing two claims from the AI-tax debate

Erosion. Two specific objections from the debate (section 1) can be put to the model directly. The first is Friedman's "the base shrinks the moment you tax it", echoed by the IMF's warning that

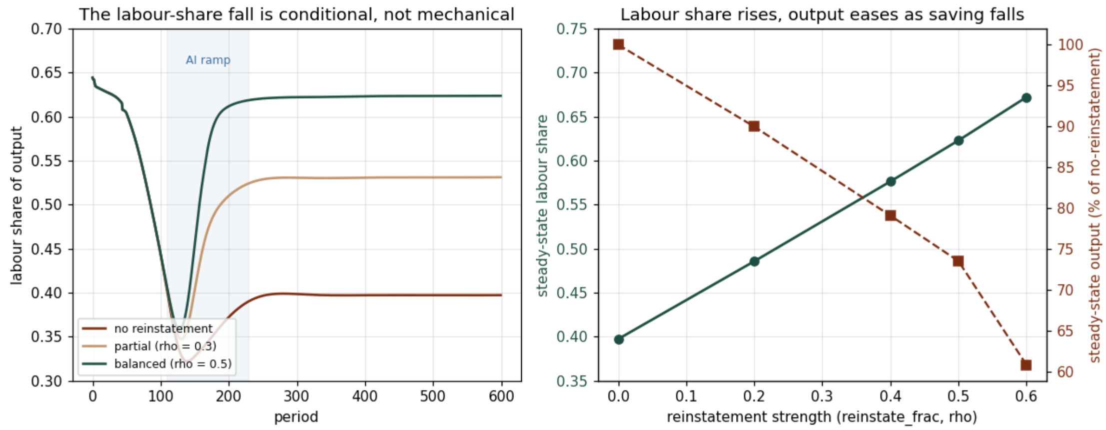

Figure 17 The reinstatement margin. Left: the labour share over time for three reinstatement strengths, with the AI ramp shaded; the share dips during automation and recovers to a level set by how many new tasks offset it, so the fall is not mechanical. Right: in the mature economy the labour share rises with reinstatement (teal, from about 0.40 to 0.67) while output eases (sienna) because the aggregate saving rate falls as income moves to labour. This is the first half of the labour-market extension; turning displacement into measured unemployment, rather than a lower wage share, is the next step.

taxing AI "could stifle productivity". Sweeping the digital levy from zero to forty per cent, with and without the AI-supply response, the response does erode the base and does cost output: with it switched on, steady-state output falls to about seventy-eight per cent of the untaxed level, and revenue comes in roughly five to twenty-five per cent below the no-response case. But revenue still rises with the rate throughout, it does not peak and turn down. So the model supports the IMF's "there is a cost" while rejecting the strong form of Friedman's claim: in the credible range the rent is durable enough that reaching it still pays, at an output cost that has to be weighed rather than a self-defeating tax. The qualifier is that the per-period deadweight is capped (section 6I), so this rules out a revenue-maximising peak only within that range, not in general.

Territoriality. The second is the territorial objection, Friedman's "a provider-level tax is a subsidy for foreign inference" and Luckey's "it just makes foreign models more attractive". Holding the levy fixed and moving the AI compute offshore (the share located at home, s_home, from 1.0 down to 0.2), the host's take is flat: the digital levy stays at about 3.7 per cent of output and the withholding at about 3.9 per cent at every server location, while the corporate base is empty throughout. The reason is that a levy on AI value recognised, and rent repatriated, within the host does not depend on where the servers physically sit, whereas a levy on the provider or the compute does. The territorial objection is therefore really an argument about the choice of base: it bites for a United States provider-level token tax (the home of the providers) and not for a rent-importing host taxing recognised value. The flatness reflects how a recognised-value base is defined, which is exactly the point rather than an artefact.

For Britain concretely: a UK digital levy on the AI revenue an OpenAI or an Anthropic earns in Britain still pays even as the firm trims UK deployment in response (the erosion is real but the take keeps rising), and it collects the same whether those models are served from data centres in Oregon or in London (the territorial neutrality). The contrast is with a United States per-token tax on the same firms, which would miss the value consumed in Britain entirely. This is why the territorial objection, decisive for Washington, does not transfer to a rent-importing host like the UK.

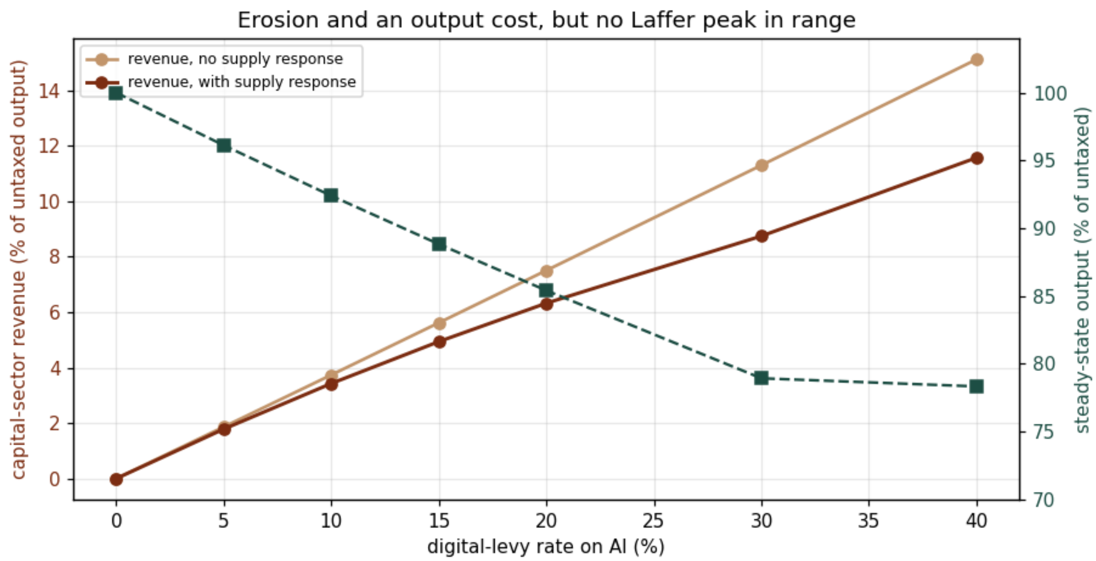

Figure 18 Testing "the base shrinks the moment you tax it". Capital-sector revenue against the digital-levy rate, without the AI-supply response (sand) and with it (sienna); the widening gap is the erosion. Output with the response (teal, right axis) falls to about seventy-eight per cent of untaxed. Revenue still rises monotonically with the rate, so within this range the levy is costly but not self-defeating.

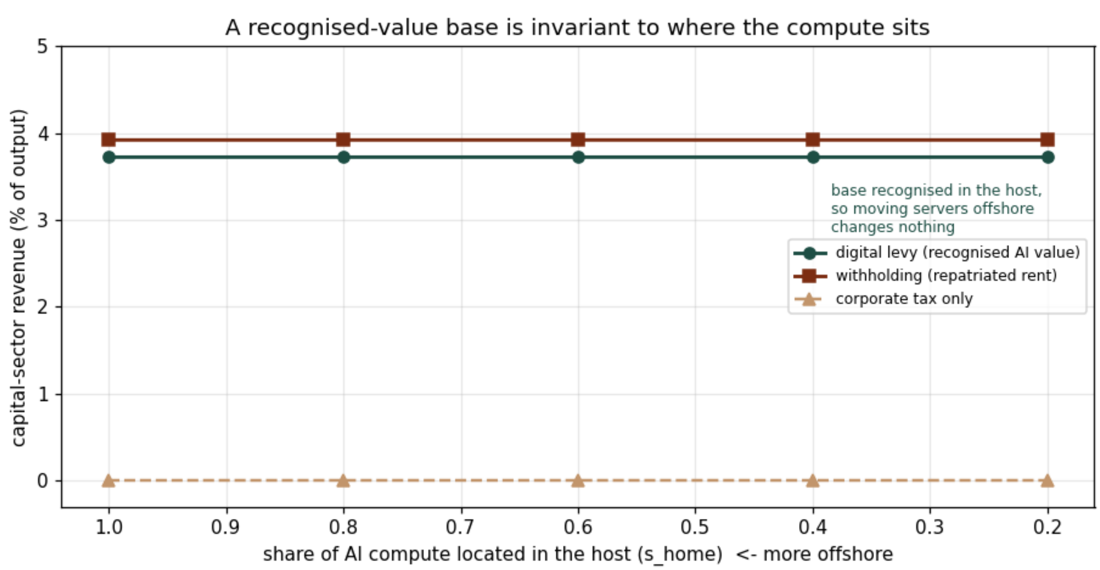

Figure 19 Testing "a domestic tax just sends the base offshore". Capital-sector revenue as the AI compute moves from fully home-located to mostly offshore (right to left). A digital levy on recognised AI value (teal) and a withholding on the repatriated rent (sienna) are invariant to compute location; only a base tied to where the servers sit would fall. For a rent-importing host the territorial objection does not bite.

6.15 L · Unemployment from automation, and funding the safety net

Employment. The reinstatement subsection moved the labour-share fall off its mechanical path; this one turns part of that fall into measured unemployment rather than a uniformly lower wage. The device is an extensive-margin overlay: production and accumulation are untouched (the displaced tasks are done by capital, which is why output holds), but a fraction of the labour displaced by automation, net of reinstatement, is recorded as out of work rather than as everyone earning less, with the displacement falling on the lowest-skill workers in each cluster first (the entry-level pattern the debate describes). The unchanged wage bill is then concentrated on those still employed, and the out-of-work receive a state-funded benefit. Because it is a relabelling of the existing labour share plus a transfer, the deposit and net-worth invariants hold to machine precision.

Two results follow. First, whether automation produces mass unemployment or a manageable rate is not a property of the technology but of the balance between displacement and the creation of new tasks. With no reinstatement and a high pass-through the rate runs toward half the workforce; with balanced reinstatement and a moderate pass-through it settles in the ten-to-twenty per cent band that Amodei warned of, after a larger transition spike. The model cannot say which case is realistic (the pass-through is a parameter, not an estimate), but it shows the optimists and the pessimists are arguing about that balance, not about whether the technology is powerful. Second, the safety net the proposals call for (McMorrow's reinvestment fund, Weinberg's lockbox) is affordable only if its funding reaches the rent: leaning on the eroding income-tax base alone, public net worth settles low, while adding the digital levy and the withholding funds the transfers and rebuilds the sovereign fund to over twice output. The instrument debate and the worker-protection debate are therefore the same debate.

For Britain concretely: if AI displaces UK cognitive roles (the entry-level analyst, the junior drafter) faster than new roles appear, the displaced need a benefit, and the UK income-tax base is itself shrinking as those wages disappear. The only funding that keeps pace is a levy reaching the AI rent that OpenAI or Anthropic books out of Britain: the same digital levy and withholding that reach the rent for revenue also pay for the safety net and rebuild the Treasury's balance sheet. A British retraining or reinvestment fund, in other words, is affordable precisely to the extent that Britain taxes the rent the automation creates.

6.16 M · Calibration and sensitivity

Calibration. The mechanisms added across the open-economy and labour-market work introduce six parameters. None is estimated for this market, which has no history to estimate from, but each can be given a literature-anchored central value and a plausible range, so the results below are read as a disciplined range rather than a point. The table gathers the anchors in one place.

Parameter	Central	Range	Anchor
AI rent size (mu_ frac)	0.25	0.15–0.40	Upper-tail markup estimates (De Loecker, Eeckhout & Unger 2020); the ranking of instruments is robust to it (Figure 13).

Parameter	Central	Range	Anchor
Repatriation share (phi)	0.50	0.30–0.70	Reinvested earnings are about 40–60% of FDI income for mature hosts, so the repatriated complement is near a half; the AI rent is a licence fee (a distributed flow), so a half is conservative.
AI-supply elasticity (eta)	0.50	0.25–1.0	User-cost elasticity of investment of about −0.5 to −1.0 (Hassett & Hubbard 2002); firm-level estimates cluster −0.25 to −0.75.
Reinstatement (rho)	0.50	0.30–0.70	Acemoglu & Restrepo (2019): displacement and reinstatement were roughly balanced before 1987 (stable labour share) and reinstatement was weaker after; a half is the recent regime.
Unemployment pass-through	0.30	0.10–0.50	Acemoglu & Restrepo (2020) find automation lowers both employment and wages, so the extensive-margin share is intermediate. The least-anchored parameter.
Benefit replacement rate	0.50	0.37–0.58	OECD net replacement rate is 58% in the initial spell and 37% long-term for a single worker at the average wage (Society at a Glance 2024).

A seventh parameter, the UBI labour-supply response, is set small: the OpenResearch guaranteed-income trial (Vivalt et al. 2024) found a one-thousand-dollar monthly transfer cut labour-force participation by two to four percentage points, mostly on the extensive margin, consistent with the earlier negative-income-tax experiments, so the model uses a modest value rather than the large response that would otherwise dominate.

Sensitivity. Running the fully-instrumented economy (the toolkit, reinstatement and unemployment all on) and moving each parameter across its range, two things stand out. First, a basic sanity check passes: each outcome's dominant driver is the parameter it should depend on, ownership on the repatriation share, revenue on the rent size, the labour share on reinstatement, and unemployment on the pass-through. Second, the results sort cleanly into robust and parameter-sensitive. Revenue and the labour share move modestly relative to their levels, and the qualitative conclusions (which lever reaches the rent, the labour-share fall being conditional) hold throughout. Foreign ownership and especially unemployment are the sensitive ones: unemployment swings by more than twenty points across the pass-through range, which is both the least-anchored and the most consequential parameter. The honest reading is that the paper's qualitative claims are robust, while the precise ownership and unemployment magnitudes should be taken as ranges set by parameters that are plausibly bounded but not estimated.

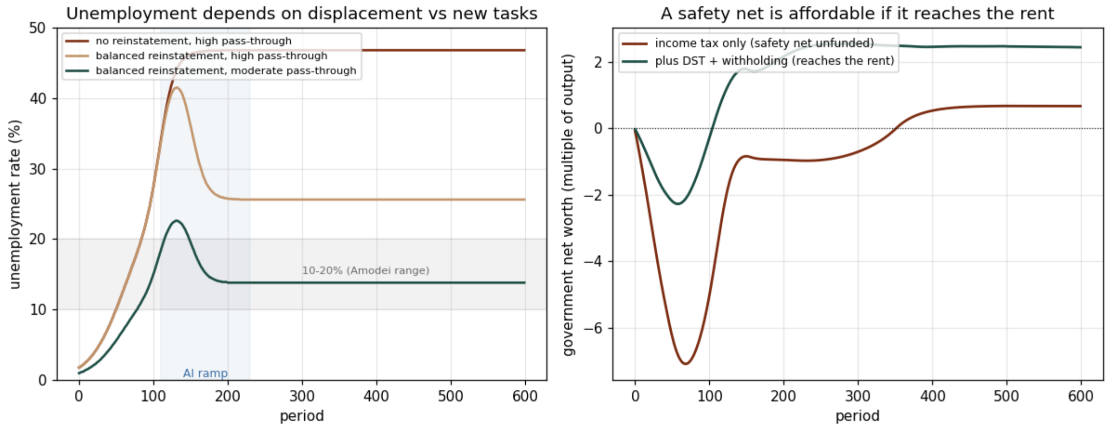

Figure 20 Left: the unemployment rate over time under three displacement-versus-reinstatement balances; balanced reinstatement with a moderate pass-through settles in the 10 to 20 per cent band (shaded), while weak reinstatement runs far higher. Right: government net worth over time when the safety net is funded by income tax alone (sienna) versus when a digital levy and a withholding also reach the rent (teal); only the rent-reaching case rebuilds the public balance sheet. Production and output are identical across these cases, the overlay is on the extensive margin only.

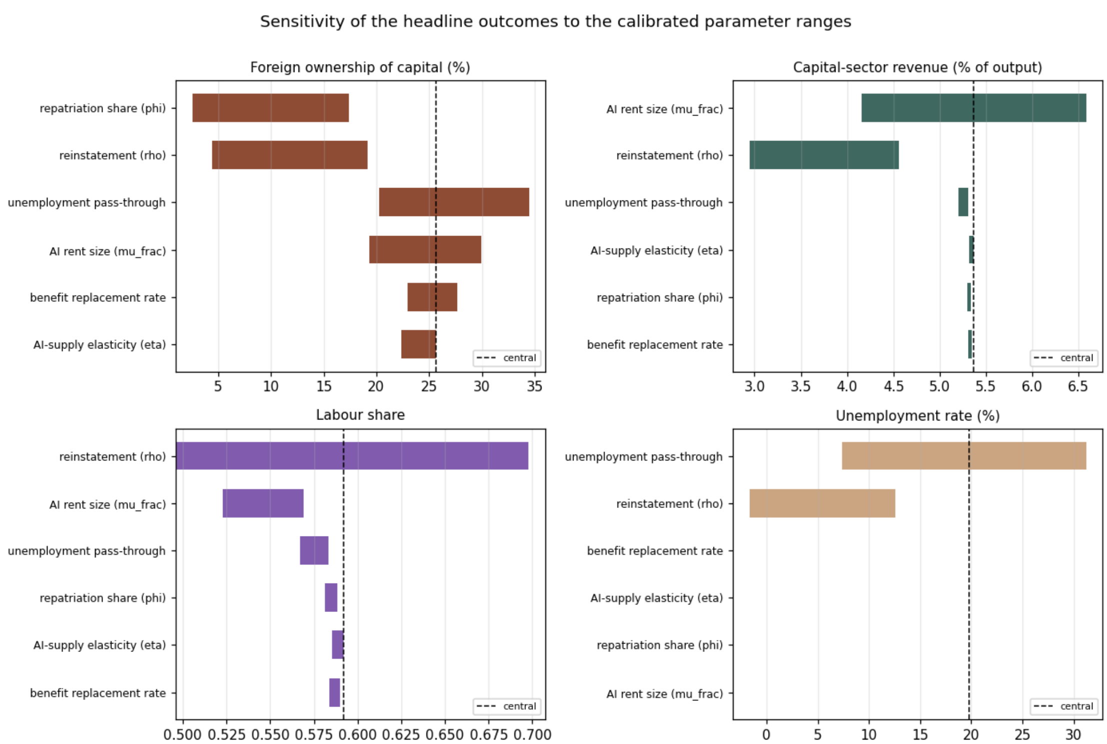

Figure 21 Sensitivity of the headline outcomes to the calibrated parameter ranges. Each bar is the swing in an outcome as one parameter spans its range (others held at the centre, dashed line); bars are sorted by size. Revenue and the labour share are comparatively robust; foreign ownership and unemployment lean on the repatriation share and the pass-through respectively, which is why those magnitudes are reported as ranges.

6.17 N · The optimising foreign owner

Strategic owner. With the open-economy and labour-market mechanisms calibrated, a further extension makes the foreign owner strategic. The territoriality test (subsection K, Figure 19) held the owner passive and found the host's take invariant to where the compute sits. That is the right answer to the wrong version of the objection. The sharper question is not where the servers are but where the rent is *booked*: an owner facing a levy on the recognised rent can move the recognition itself to a low-tax entity, the standard profit-shifting move. The model now lets a share of the rent recognition shift offshore, rising with the host's wedge and capped because some of the rent is tied to recognised domestic use, so the shifted portion escapes both the digital levy and the withholding. With the elasticity set to zero this is the non-strategic owner of the earlier sections exactly.

Two things follow (Figure 22). The host's take now bends below the non-strategic line, with the gap, the rent booked offshore, widening as the rate rises: the reach is real but bounded by how cheaply the owner can shift, calibrated here to a profit-shifting semi-elasticity of around 0.8 (Heckemeyer and Overesch 2017). And a more mobile owner both pays less and owns more: at the full-toolkit wedge, raising the owner's mobility cuts the host's revenue from about 7.6 to 5.3 per cent of output and lifts foreign ownership from about 34 to 48 per cent, because the rent that escapes the levy stays with the owner to reinvest. The honest refinement of the territoriality result is therefore this: a rent-importing host has more reach than the United-States-centred debate assumes, because moving the servers does not erode a recognised-value base, but that reach is not unconditional, because moving where the value is booked does. The policy implication is less "the base cannot move" than "the base moves through transfer pricing, not server location", which points at recognition and anti-avoidance rules rather than at where the compute is hosted.

For Britain concretely: the AI owner (an OpenAI or an Anthropic, resident in the United States or China) cannot dodge a UK levy on recognised AI value by serving its model from Oregon instead of London, which is the reassuring half of the result. But it can book the UK licence fee through a low-tax intermediary, the familiar transfer-pricing route, and that does shrink what the UK Treasury collects. The reading for Britain is that a digital levy has to be written against recognised UK value and paired with transfer-pricing and recognition rules, otherwise the base leaks through the accounting even though it cannot leak through the data-centre map.

6.18 O · The contestable frontier: competition as an alternative to tax

Competition. The rent has so far been a fixed share of cognitive value added. That is right only if the frontier stays monopolistic. A further extension makes the markup respond to contestability: open-weight catch-up and more providers price part of the rent away, so the effective rent is the markup net of competition. With contestability at zero this is the monopolistic frontier of the earlier sections; at one the rent is fully commoditised. This matters because it turns competition policy into a lever that sits alongside taxation, and the two behave very differently.

As contestability rises the rent shrinks and, with it, foreign ownership: the rent was the durable surplus the foreign owner accumulated, so competing it away takes ownership from about 69 per cent toward zero (Figure 23, left). The labour share barely moves, because the rent was peeled from both the cognitive wage and the domestic AI capital income, so returning it benefits both factors rather than labour alone, a useful corrective to the intuition that killing the rent straightforwardly reflates wages. The instrument contrast is the point (Figure 23, right): competition and taxation both cut foreign ownership, but they are opposites for the public finances. Taxation captures the

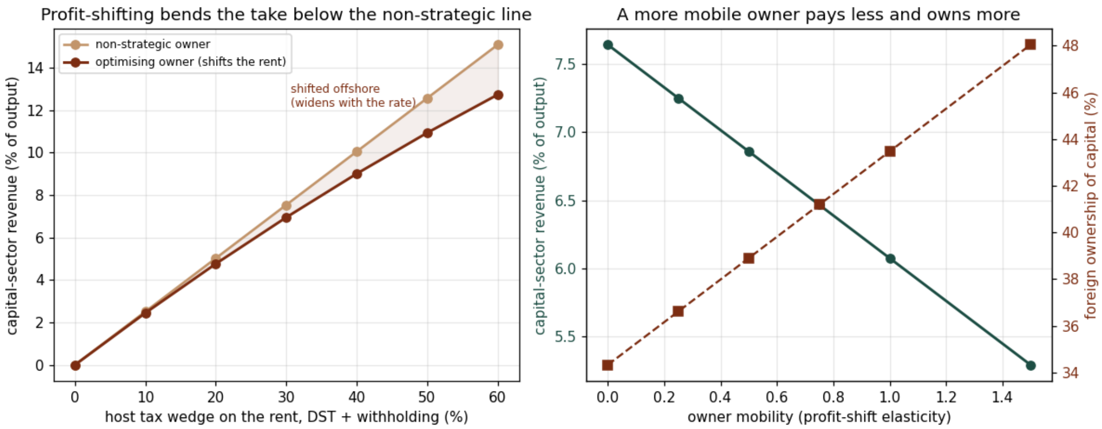

Figure 22 The optimising foreign owner. Left: the host's take against its tax wedge on the rent, for a non-strategic owner (sand) and one that shifts the rent recognition offshore (sienna); the shaded gap is the rent booked elsewhere, widening with the rate. Right: at the full-toolkit wedge, as the owner's mobility rises, the host's revenue falls (teal) and foreign ownership rises (sienna), since the shifted rent stays with the owner to reinvest. Reach survives server relocation but is bounded by profit-shifting of the recognition.

rent, so revenue climbs as ownership falls; competition destroys the rent, so there is nothing left to tax and revenue stays at its low floor. They are substitutes for the ownership and distributional goal and opposites for the fiscal one. This is the model's way of stating a choice the debate gestures at: the IMF and Brookings preference for reviewing IP and innovation incentives, and Friedman's expectation that prices will be competed down anyway, point toward the competition lever, which is first-best if the concern is that the rent should not exist; a levy is the instrument if the concern is that, while it exists, the host should capture rather than merely dissipate it. The honest limit is that contestability here is an exogenous market or policy state, not yet entry responding endogenously to the size of the rent, which would be the fuller treatment.

For Britain concretely: if open-weight models (a Llama, a DeepSeek) commoditise the frontier, the UK-derived rent of an OpenAI or an Anthropic shrinks on its own, and foreign ownership of British capital falls without the UK levying anything. But that same erosion leaves the Treasury with nothing to tax. So Britain faces a real choice of route, not just of rate: backing open models and competition serves the ownership and consumer-price goal, while a digital levy serves the fiscal goal, and only the levy puts money in the public purse.

This is not a hypothetical lever, and it may already be the live one. China's leading labs (DeepSeek, Alibaba's Qwen, Moonshot, Baidu) release their strongest models as free open weights, a strategy a US government commission report of March 2026 reads as deliberately undercutting the American providers' keep-it-behind-an-API-and-charge model and diffusing Chinese models worldwide. By early 2026 Chinese open-weight models led the most-downloaded charts, their token usage on at least one routing platform had overtaken the US models, and one venture-capital partner estimated that, among the start-ups it sees which run on open weights (perhaps a fifth to a third of its deal flow), roughly four in five build on a Chinese model; OpenAI's own move to release open weights was, on its chief executive's account, a response to the prospect that the world would otherwise be built on Chinese open source. In the model's terms this is the contestability lever pushed from the outside: if open weights commoditise the frontier, the rent erodes whether or not any host taxes it, which removes the ownership-drift problem and the tax base at the same time. This is something the model already represents rather than something it would need a new mechanism to show, and it implies a rent-importer has a second, non-fiscal route to the ownership goal (backing open models) that a country hosting the providers would be far more reluctant to take, since it would be commoditising its own champions.

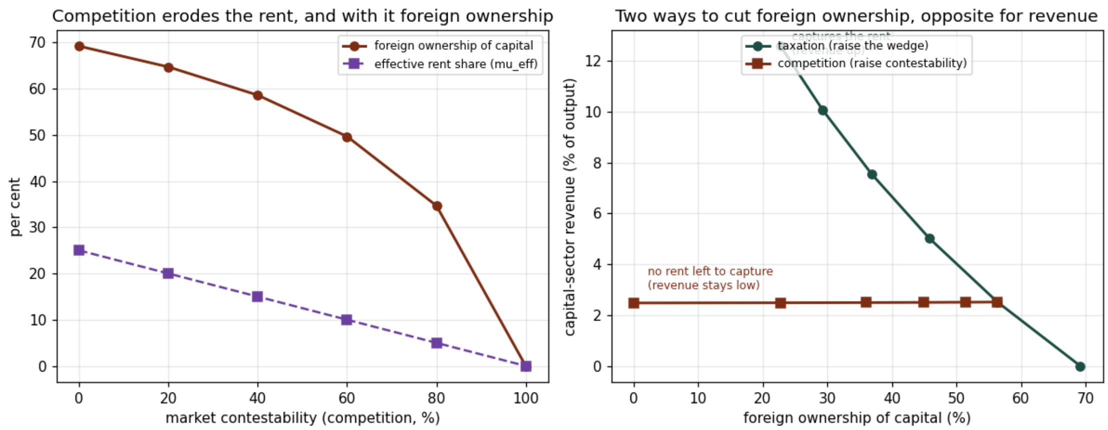

Figure 23 The contestable frontier. Left: as market contestability rises, the effective rent share (purple) falls and foreign ownership (sienna) collapses with it. Right: both taxation and competition cut foreign ownership, but taxation raises revenue as it does so (it captures the rent) while competition leaves no rent to tax (revenue stays at its floor). The levers are substitutes for ownership and opposites for revenue.

6.19 P · Valuing the rent: the capitalised IP, and why a market price adds nothing

Valuation. The paper has worked in flows: the rent per period, the tax per period, the ownership share. A natural question for the ownership story is what the rent is worth as a *stock*, capitalised, because that is the owner's true wealth and it is where any wealth-tax or asset-price argument would have to bite. This subsection capitalises the after-host-tax rent the model produces, at a required return, as a valuation overlay. It was developed and validated in a separate sandbox and is kept off the shipped model, so none of the earlier results move; and it is a reduced form, a discount rate and a premium rather than estimated ones, so the magnitudes are illustrative rather than calibrated.

The capitalised IP is worth about 2.2 years of output at the baseline rent and scales with it (Figure 24, left, purple). Because it is the foreign owner's asset and sits above the book capital stock, it lifts foreign ownership measured at market value above book, from about 69 to 75 per cent at the baseline and more at a larger rent (the shaded gap). A source levy on the rent erodes this capitalised wealth, from about 2.2 to 1.3 years of output across the wedge (Figure 24, right): a digital-services levy or a withholding does not merely take a slice of this year's rent, it lowers the present value of the owner's claim. A residence-based wealth tax, by contrast, cannot reach the IP at all, because it is foreign-held and offshore, which is the same structural reason the paper's instrument is a source levy rather than a tax on domestic wealth.

It is worth saying why a fuller, market-clearing price was not pursued. The intuition behind it, the owner bidding up the price of ownership as it accumulates, turns out to be degenerate in this economy. The domestic equity yield is slightly negative in the mature steady state, even with no rent, because the competitive return on capital is competed away by automation and the only surplus, the rent, is siphoned to the foreign owner. A Tobin's q on domestic equity therefore sits below replacement cost and *falls* as the rent rises, the opposite of a bidding-up story, while the asset that does hold the value, the rent-bearing IP, is singly held and so has no market to clear; pricing it simply returns the fundamental already shown here. The reproduction check on this attempt passed cleanly (a switch recovers the book model exactly, with the accounting invariants holding to machine precision), so this is a clean negative result rather than a failure. It is the third independent route to the same fact: in an automated economy of this kind the durable surplus is the foreign-held IP rent, not a return on the domestic capital stock, and that is precisely why a market price adds nothing and a source levy on the rent is the instrument that reaches the surplus.

For Britain concretely: the AI owner's UK-derived IP is worth a multiple of a year's rent, and it sits on a balance sheet in San Francisco or Shenzhen, out of reach of any UK wealth tax. A UK source levy, by contrast, lowers the present value of that claim directly, because it taxes the rent the claim capitalises. This is the asset-side version of the whole paper's point: the thing worth owning is offshore, so the instrument that reaches it is a tax at the British border on the value recognised there, not a tax on wealth held in Britain.

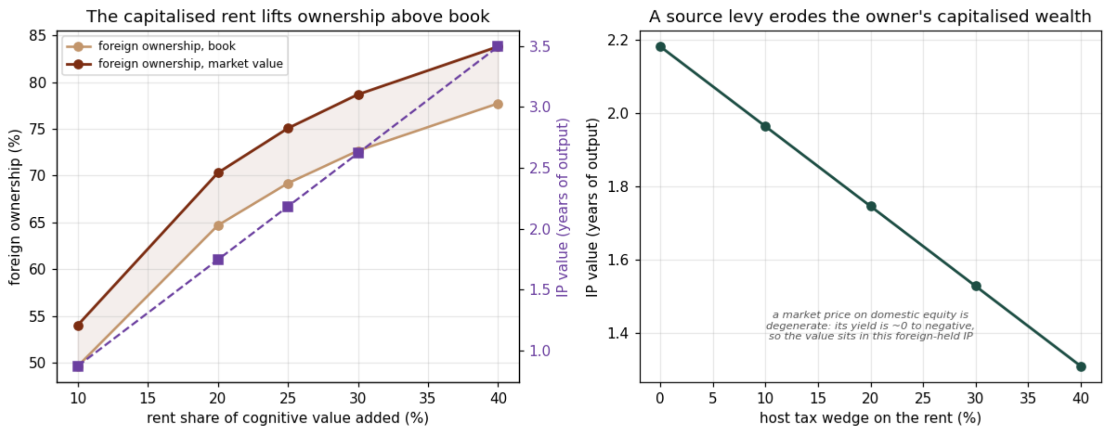

Figure 24 Valuing the rent as a stock. Left: across the rent share, the capitalised IP value in years of output (purple, right axis) and foreign ownership measured at book (sand) versus at market value (sienna); the capitalised rent lifts the owner's market-value share above book, and the gap widens with the rent. Right: a source levy on the rent erodes the present value of the owner's claim. A market price on domestic equity is degenerate here because its yield is around zero to negative, so the value sits entirely in the foreign-held IP.

6.20 Q · Two stacked foreign rents: compute and model

Stacked rents. The paper has bundled all durable foreign surplus into one IP licence rent on the cognitive cluster. In reality the compute layer earns its own rent: the accelerator design and its software ecosystem (the NVIDIA-and-CUDA case) command a margin that is mostly foreign-held and, unlike the model rent, embedded in the *price* of AI capital rather than flowing as a recurring licence fee. This extension (foreshadowed in Appendix A.2) splits the foreign owner into two rentiers: the existing IP rent, and a compute rent (`mu_compute`) peeled from the AI capital income, with its own domicile and its own reachability. The point of interest is not its size but its *form*: a withholding or a digital-services levy sits in the path of a licence flow, not a goods price, so the instruments the paper recommends do not obviously reach it.

The model bears this out. A compute rent of about six per cent of output, stacked on the IP rent, lifts steady-state foreign ownership from about 69 to about 75 per cent (Figure 28): a second foreign rent accelerates the drift of ownership abroad, exactly as the embodied-robot-IP rent did (subsection in Appendix A.2). The instrument contrast is the result. The digital-services levy and the withholding, which capture the licence rent, collect *nothing* from the compute rent, because it never appears as a cross-border fee; only a tariff on imported compute (about 1.7 per cent of output recovered here) or a usage levy on inference (about 3.7 per cent, since it falls on the whole compute bill) reaches it, and a domestic chip-maker keeps it home outright (foreign ownership falls to about 44 per cent, and to zero when both rents are domestic). Onshoring the servers does not help: with the compute on foreign-owned hardware (low `s_home`) the take is unchanged, because the rent is paid in the capital-good price wherever the box physically sits. So the stacked rent both deepens the importer's predicament and adds a surplus that the paper's own reaching instruments miss, unless the host reaches for a border or usage tax. It is worth being explicit that this reinforces the paper's conclusion rather than threatening it: a second, hardware-embedded foreign rent makes the durable surplus larger and the ownership drift faster, so the case for reaching the rent early is stronger, not weaker. What it qualifies is the *sufficiency* of the recommended instrument, not its *direction*; the digital levy and the withholding remain the right tools for the licence rent, and the compute rent simply adds a border or usage instrument alongside them. The point belongs in the limitations as future work (a fuller treatment would give the two rents independent persistence and an up-front capex form), but the direction of the headline result is untouched.

For Britain concretely: a rent-importer that taxes the model rent with a digital levy still leaks the chip rent embedded in every GPU it imports, and onshoring the data centres does not change that. Reaching it needs either an instrument at the border or a domestic stake in the compute layer, neither of which is the levy the rest of the paper recommends.

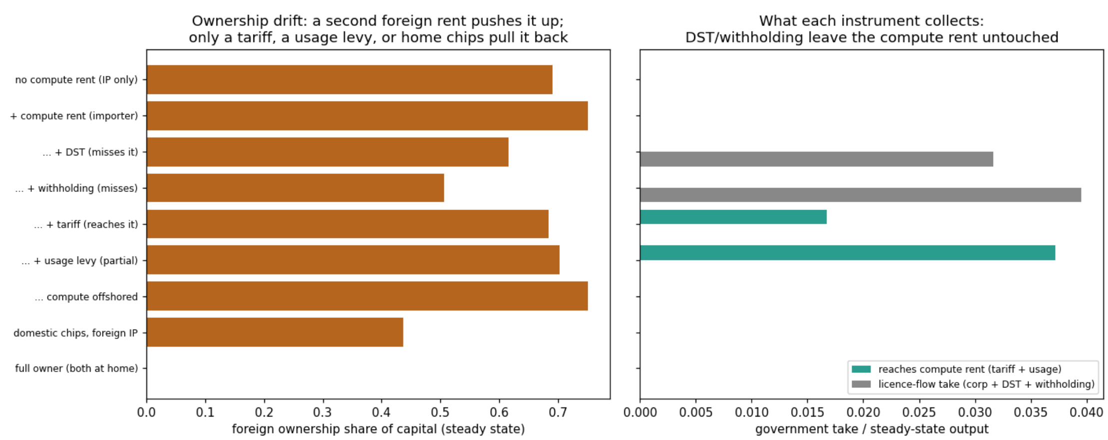

Figure 28 Two stacked foreign rents. Left: steady-state foreign ownership of capital. A compute rent stacked on the IP rent raises it; a tariff, a usage levy, or a domestic chip-maker pull it back, while offshoring the servers leaves it unchanged. Right: what each instrument collects. The digital-services levy and the withholding (grey) reach the licence rent but collect nothing from the compute rent; only the tariff and usage levy (teal) touch the goods-embedded rent.

6.21 R · Elastic labour supply and the bottleneck wage

The bottleneck wage. The model pays each labour type its marginal product with fixed supply, so as robotic capital accumulates the task it cannot do (the physical, routine bottleneck) becomes a scarce complement and its wage rises sharply: this is the model's "automation raises wages'' result. The objection is that a task hard for a machine but easy for any human need not command a high wage, because abundant labour competes it down. This extension makes the effective labour supplied in each cluster slope up in its own per-unit wage, with a reservation floor, solved as a short fixed point against the technology.

Letting the bottleneck labour be elastic competes its scarcity premium away (Figure 29). Under fixed supply the routine per-unit wage runs to roughly thirty-six times the cognitive one; with an elastic supply that ratio collapses toward three, and within the bottleneck cluster the surplus shifts to capital (its labour share falls from about 0.79 to about 0.77). The honest qualification is that this is a result about the bottleneck *wage*, not the economy-wide labour share: because the labour-intensive cluster expands as labour floods in, the aggregate labour share need not fall and may even rise slightly. So the experiment does not show that automation lowers the labour share; it shows that the dramatic bottleneck-wage spike in the headline model is an artefact of inelastic supply, and that the reassurance "the workers doing what AI cannot will be paid handsomely'' is fragile once that labour is abundant.

The general lesson is that the wage outcome turns entirely on what the remaining human task is, and the headline result is one of three cases, not the generic one. Where the bottleneck labour is paid its marginal product in genuinely *fixed* supply, as in the headline model, its wage rises in level even as the labour share falls; this is the comforting reading. But that is the knife-edge. Where the task is one *any* worker can do, the elastic-supply case above shows the scarcity premium competes away and the surplus accrues to capital rather than to labour. And where it is instead a genuinely *scarce*, hard-to-acquire skill that only a few hold, the premium survives but is captured by a narrow wage elite rather than by labour at large. So "automation raises wages'' holds broadly only under fixed supply; the two generic outcomes are a return to capital or a within-labour concentration, both of which sharpen rather than soften the distributional story the rest of the paper tells. The level claim should be read with this qualification throughout, and it is foregrounded again among the caveats.

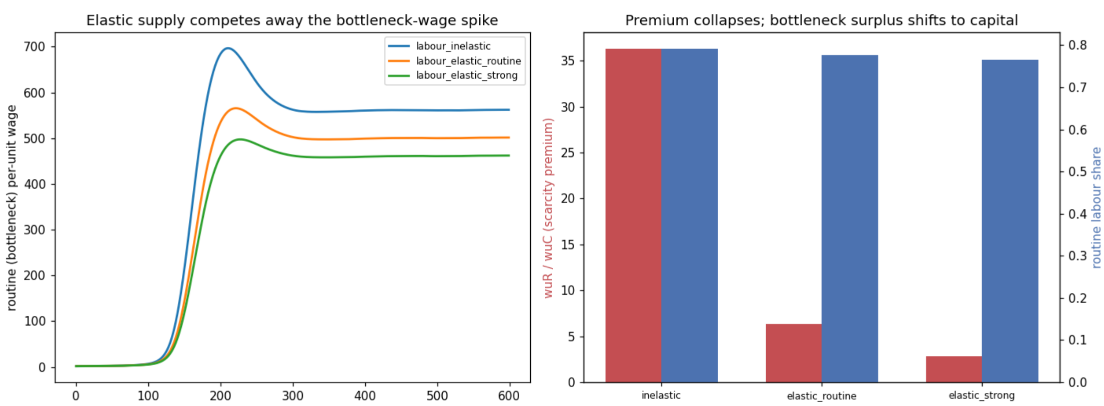

Figure 29 Elastic labour supply and the bottleneck wage. Left: the routine (bottleneck) per-unit wage path; an elastic supply competes away the spike that fixed supply produces. Right: the routine/cognitive wage premium collapses as supply is made elastic, and the bottleneck cluster's labour share falls (the surplus moves to capital), while the aggregate labour share need not.

6.22 S · Endogenous, cost-driven automation

Paying for itself. Automation has so far been an exogenous logistic ramp. The MIT "Beyond AI Exposure'' result is that only cost-effective tasks are automated, and gradually. This extension keeps the logistic as the technical frontier but gates it by cost: a task is automated only when the machine is cheaper than the human wage for it, with the machine cost starting above the wage and declining at a calibrated rate, and rising bottleneck wages opening the gate further (the feedback).

The transition still arrives, but its pace is set by the cost decline (Figure 30). Where the exogenous frontier reaches near-full automation by about period 125, a slow cost decline delays the realised transition to about period 350 and leaves it partial for long stretches; a middle (MIT-baseline) decline brings it to about period 195; a fast decline returns it almost to the frontier timing; and AI-as-a-service scale economies pull it forward. Crucially it does arrive in every case, because once a little automation raises the bottleneck wage the gate opens wider, so the result is a question of *how fast and how far*, not *whether*. The timing of the distributional shift (the rise in foreign ownership and in wealth concentration) tracks the cost-decline speed directly, which quantifies the "premise might not hold yet '' caveat without overturning it: a realistic, cost-paced rollout still delivers a substantial transition over a couple of decades.

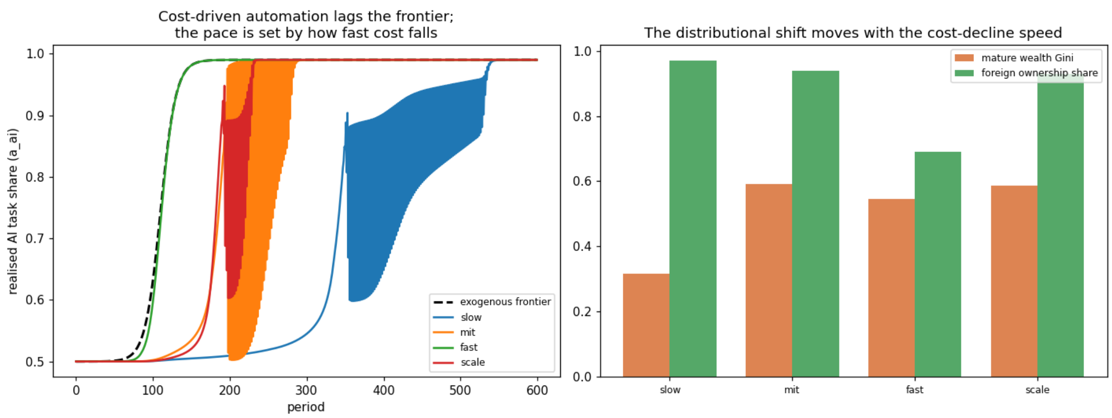

Figure 30 Endogenous, cost-driven automation. Left: realised AI task share against the exogenous frontier (dashed). Cost-gated automation lags the frontier, and the lag is governed by how fast the machine cost falls. Right: the mature wealth Gini and the foreign-ownership share move with the cost-decline speed, so a slower rollout postpones the distributional shift rather than preventing it.

7 Further results: condensation speed, leakage, and the cost of waiting

Three further dynamics complete the picture. Each is reported in the same way: what the experiment does, what the result is, and why it matters. They concern the dynamics of the transition (how fast inequality concentrates, how foreign ownership drains the fiscal base, and how the timing of intervention matters), and they are robust across the model's generations; two of the figures are retained from earlier runs of the programme and illustrate the mechanism rather than re-stating the section 6 quantities.

7.1 Speed of condensation rises with automation

What we do: vary the depth of automation and track how fast the wealth distribution concentrates when concentration is coupled to the capital share. *Result:* deeper automation produces both higher terminal inequality and faster condensation toward it. *Why it matters:* the speed, not just the level, is policy-relevant, because a faster condensation leaves a shorter window in which intervention is cheap, which is the theme the timing result below makes precise.

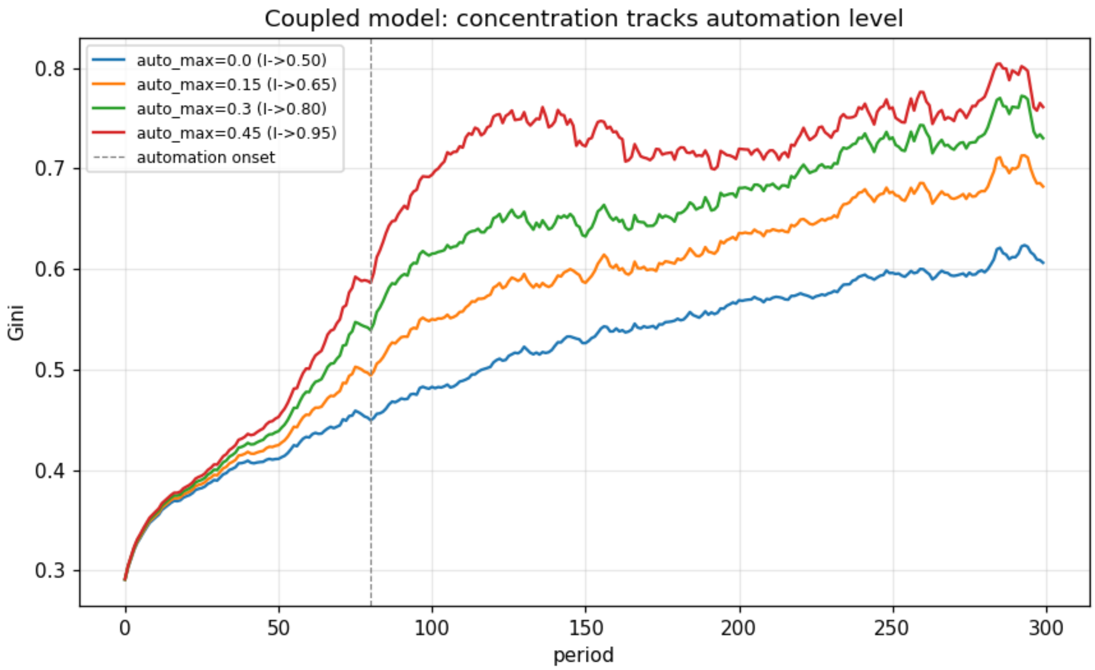

Figure 10 Deeper automation produces both higher and faster wealth condensation when concentration is coupled to the capital share.

7.2 Foreign ownership leaks wealth and erodes the fiscal base

What we do: start the rest of the world with a foreign equity stake and let the ownership share evolve under the saving and tax rules, tracking repatriated income and government net worth. *Result:* the dedicated foreign-ownership scenario starts the rest of the world at a 60% equity stake but does not pin it. Because new issuance is subscribed by whoever is saving, and the big domestic saver under a 30% income tax is the state, the foreign stake dilutes over the horizon and is, in effect, gradually transferred to the domestic public sector: by the terminal period the scenario looks like the income-tax-plus-UBI economy with a sovereign equity fund. *Why it matters:* the leakage of repatriated profit while the stake lasts still erodes the fiscal base during the transition, which is exactly the drain the two-channel AI rent makes permanent unless an instrument reaches it (section 6G). Sustaining a constant foreign share would require modelling persistent FDI inflows, whose domestic counterpart is an ever-growing gross creditor position under r>g; that is flagged as future work rather than imposed, since imposing it introduces a spurious balance-sheet explosion.

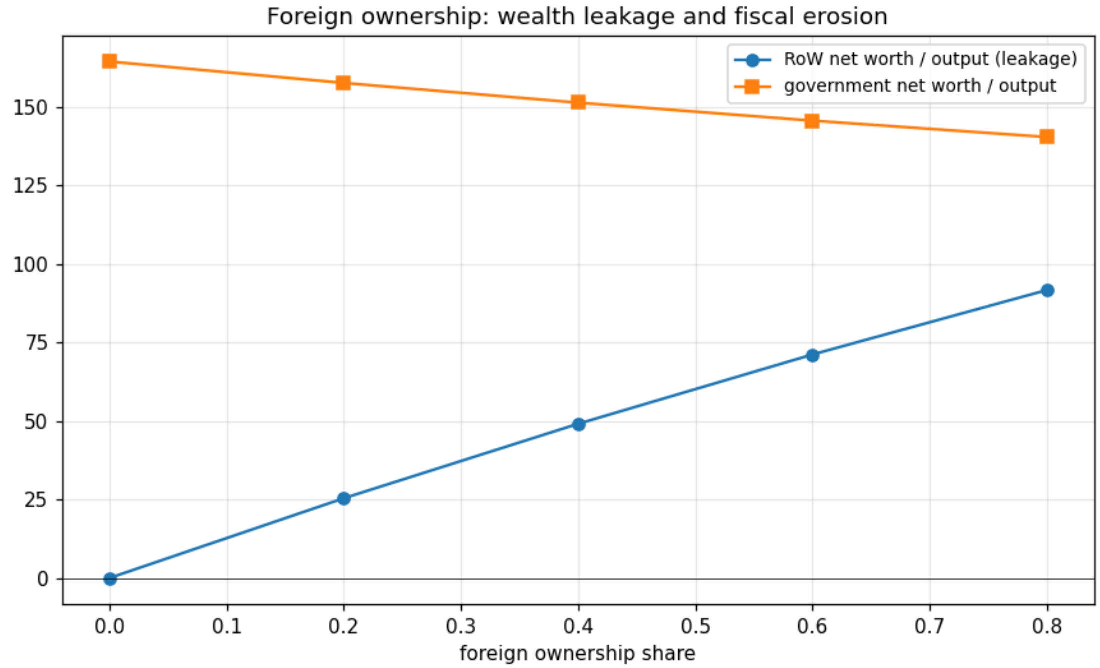

Figure 11 Repatriated wealth rises and government net worth falls with the foreign ownership share. The model makes this share endogenous to the tax (experiment B), and the two-channel extension makes it endogenous to the AI rent (Figure 8).

7.3 Intervention timing exhibits hysteresis

What we do: introduce the same redistributive tax at progressively later dates and compare terminal inequality. *Result:* the later the intervention, the higher the terminal inequality, and the penalty grows more than linearly with delay. *Why it matters:* once concentration has compounded it is expensive to reverse, so the timing of a policy is itself a lever; combined with the first result (automation speeds condensation up), it implies the window for cheap intervention narrows precisely as automation deepens.

8 Policy implications for the UK

This section reads the results as conditional policy implications. The model is positive, not normative, so it cannot say what the United Kingdom *should* want. If the objectives are taken to be a long-run stable economy and contained inequality, which is the assumption made here, then the experiments point fairly consistently in one direction. The magnitudes are illustrative (the AI rent is a calibrated markup, not an estimate) and the ownership results are conditional on the no-goods-trade closure, so what follows is about the direction and ranking of instruments rather than precise numbers.

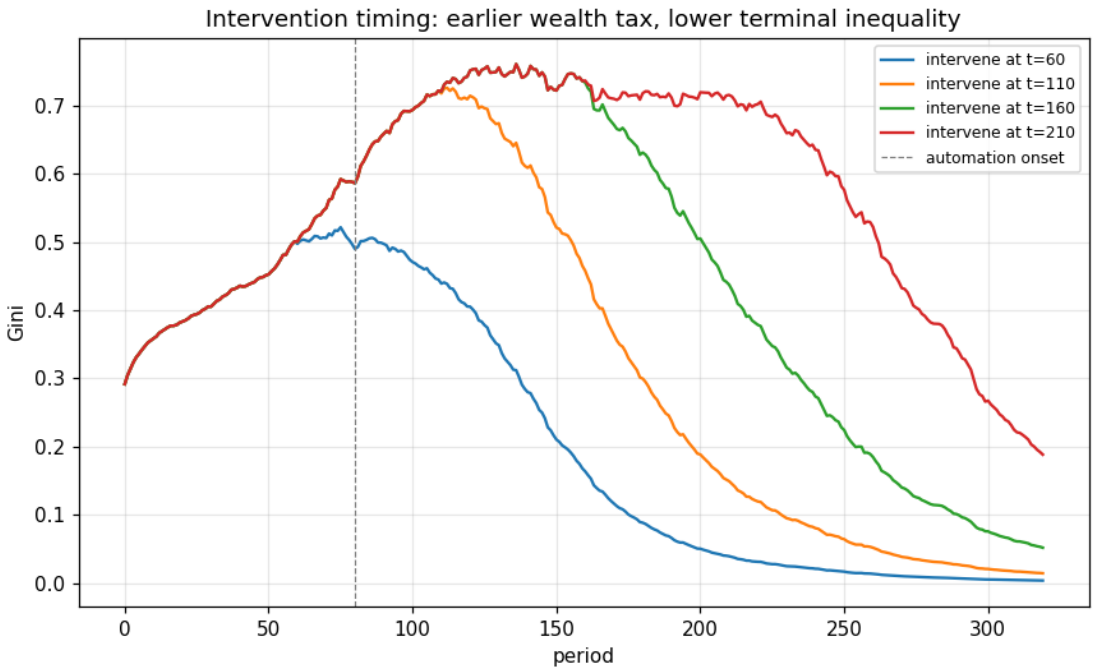

Figure 12 The later a wealth tax is introduced, the higher the terminal inequality; the cost of waiting is convex.

8.1 Two regimes, two policy menus

The single most important determinant of the right policy is one the debate usually leaves implicit: whether the AI is owned at home or abroad. The two regimes call for almost opposite instruments, which is why the same proposal can look right to an American commentator and wrong to a British one.

When the owner is domestic (the United States), the rent stays in the country and the problem is its concentration among domestic owners. The instruments that reach it are the ordinary domestic ones: progressive capital, dividend and capital-gains taxes, a wealth tax (which works here precisely because the owner is resident and the IP onshore), and competition policy. The binding constraint is competitiveness, since taxing a domestic champion risks pushing its activity offshore, which is where the territoriality objection genuinely bites. The benchmark in section 6 shows the shape of it, with a domestic wealth tax compressing the home inequality the rent creates.

When the owner is foreign (the United Kingdom, with a United States or Chinese owner), the rent leaves as a licence fee and the problem is lost ownership and an eroding base. Now the domestic levers fail: a wealth tax cannot reach offshore IP, and ordinary corporate tax misses a deductible fee. What works instead is a source levy, a digital-services tax or a withholding written against value recognised at home, and the binding constraint is not server location (the take is invariant to it) but where the rent is booked, so the levy must be paired with anti-avoidance rules. This is the case the rest of this section is about.

The asymmetry is the paper's organising point: the same AI rent calls for domestic progressive taxation in the country that owns it and a source levy in the country that imports it, and the rent-importer has more reach against relocation, not less, than the provider's home. The remainder of this section develops the foreign-owner menu, the harder and less familiar one.

8.2 Reading the model against the AI-tax proposals

The live proposals (section 1) are, at bottom, arguments about the tax base, and the two-channel model is a way to ask which base reaches the surplus. Four mappings stand out, and they cut for and against different parts of the debate.

The base matters more than the label, and the durable surplus is the rent. The model's clearest result is that the lasting surplus from automation is the mobile AI IP rent, a markup on the cognitive cluster, not the competitive return on the physical machines. A levy that tracks AI revenue (the digital levy here) or the repatriated rent (the withholding) reaches it; ordinary corporate tax reaches an almost empty base, and a robot tax lands on the low-rent physical channel. If robots themselves carry an embodied-IP rent (the model running them), an off-default lever confirms it is reached by the same source levy and missed by the hardware robot tax, so this ranking is unchanged and the durable surplus is simply larger (Appendix A.2). Read against the debate, this is why a per-token charge is a weak instrument: a token tracks an internal accounting unit, not the rent, which is Friedman's neutrality objection in the model's own terms. The base that bites is a share of AI value, or of the rent as it is recognised, which is closer to Friedman's own suggested alternatives (a gross-receipts or inference-revenue base) and to Amodei's mooted revenue levy than to a token count.

The base does shrink when taxed, but the alternatives reach less. Friedman's strongest point is that the base erodes the moment it is taxed. The model now contains exactly that channel (the AI-supply response, section 6I): taxing the rent is not free, and a high enough rate lowers output. But the same model shows that the instruments which avoid this objection, ordinary corporate tax and the robot tax, reach almost none of the durable surplus. The honest conclusion is not that taxing the rent is costless, it is that the choice is among imperfect instruments, and a revenue-or-rent base dominates a token or physical-capital base on reach, at a cost that has to be weighed rather than wished away.

Territoriality bites for the provider's home, far less for the rent's host. The territorial objection (Luckey, and Friedman's fourth point), that a domestic levy just sends activity to untaxed foreign providers, is decisive when the taxing country is the home of the providers: a United States per-token tax on United States firms captures none of the value that routes through non-US inference. The model's open-economy and compute-location results say something more specific for a host that imports the rent. What such a host can reach does not depend on where the servers sit (the s_home result: onshoring compute barely moves the steady state) but on whether it taxes the value recognised, and the rent repatriated, within its borders. A revenue or withholding levy on AI value consumed in the jurisdiction is therefore much harder to route around than a provider-level token tax, because the consumption and the licence fee are in the host whatever the provider's domicile. Luckey is thus right for the United States and wrong as a general claim, an asymmetry a UK or EU policymaker should hold onto. The reach is bounded rather than unconditional, though: an optimising owner can still move where the rent is booked, even if not where the servers sit (subsection N), so the binding margin is transfer pricing and recognition rules, not compute location.

The worker-fund question is the rebate-versus-bank choice. McMorrow earmarks the proceeds for an AI Workforce Reinvestment Fund and retraining; Weinberg's lockbox does the same for displaced workers; Warren would bank an energy-excise take for public goods. The model frames this as the one distributional choice it cannot make for the policymaker (see the package below): a rebate or worker fund delivers the lowest measured inequality but builds no public saving, while banking the levy in a sovereign fund rebuilds public net worth and re-domesticates ownership but does less for near-term inequality. The reinstatement result (section 6J) adds a wrinkle: the displacement these funds are meant to cushion is not mechanical, since how far the labour share falls depends on whether new tasks are created, so a fund that finances genuinely new roles (McMorrow's apprenticeship framing) acts on the same margin the model identifies, not only on the transfer.

None of this settles whether to tax AI. It does suggest the productive disagreement is not "tokens, yes or no" but the choice of base and the question of ownership, and that a rent-importing economy has more reach than the United-States-centred debate assumes.

8.3 The proposals, one by one

The mappings above are organised by theme. It is worth also walking the named proposals one at a time, because the model has something specific to say about most of them, and is honestly not equipped to judge a few, and the distinction is itself a result.

Take the token-base proposals first. Mark Cuban's per-token charge, and the "Token Tax" that Mallory McMorrow built into a worker-protection plan, both make the unit of tax the token a model emits. The model's verdict on that base is unambiguous and lines up with Dave Friedman's critique: a token is an internal accounting unit, not a unit of value, and the durable surplus here is a rent earned on the model and the brand behind it, not a quantity of tokens. A levy on recognised AI value, or a withholding on the rent as it crosses the border, reaches that surplus; a token count does not, because it tracks the wrong thing and a provider can redefine it. Where the model has to stop short is exactly where Friedman's case is most mechanical. It contains no tokens and no tokenizer, so it cannot reproduce his argument that the base is endogenous to the taxpayer, nor his observation that per-token prices have fallen roughly two-hundredfold a year, which would make any fixed per-token rate quickly either confiscatory or trivial. The model reaches his conclusion, do not tax tokens, by a different route, and should be read as agreeing with it rather than as having tested those two particular failure modes.

Dario Amodei enters the debate twice, and the two should be kept apart. His widely-reported warning is about displacement, that AI could remove a large share of entry-level white-collar work and push unemployment into double digits. His own tax suggestion, by contrast, is a modest levy of around three per cent on model revenue, not on tokens. The model is favourable to that revenue framing: a revenue base sits close to the base that actually bites, because it scales with the value the rent is extracted from in a way a token count never could. On the displacement warning, the labour-market block reproduces the ten-to-twenty per cent unemployment band, but as a conditional result, with the outcome turning on whether new tasks appear as fast as old ones vanish rather than on how capable the technology is. The limitation here is the sharpest in the paper. The speed at which displacement passes through to joblessness is the least anchored parameter in the model and the one the outcome is most sensitive to, which is why that band is a range and not a forecast, and why the revenue-maximising rate for a levy like Amodei's is not something the model can pin down.

The robot tax, associated with Bill Gates and defended by economists such as Edmund Phelps, fares better on administrability than on aim. Because robots are physical and cannot be spirited abroad, a tax on them is clean to levy and hard to avoid, and the model confirms it lands without leakage. But it lands on the wrong channel. The robotic side earns an ordinary competitive return that automation drives towards zero, so it carries very little of the lasting surplus, while the rent sits on the AI side and leaves as a licence fee. A robot tax is therefore a sensible complement on the physical channel, not the centrepiece. The caveat is that this ranking depends on the split the model draws between the AI and robotic clusters, which is a calibrated assumption, and on robots themselves not relocating.

Palmer Luckey's territorial objection, that a domestic AI tax simply makes untaxed foreign models more attractive and invites a surveillance apparatus to enforce it, is the claim the model engages most directly, and the answer is strikingly asymmetric. For a country that hosts the providers, such as the United States, the objection has force. For a country that imports the rent, such as the United Kingdom, it does not: the model's revenue is invariant to where the compute is physically located, because a levy on value recognised at home does not care whether the servers sit in London or Oregon. The qualification, which an experiment added rather than assumed, is that the reach is bounded not by server location but by where the rent is booked. A strategic owner can still route the licence fee through a low-tax entity, so the levy has to be written against recognised domestic value and paired with transfer-pricing and anti-avoidance rules. On Luckey's enforcement-and-surveillance concern the model is silent, and the profit-shifting it does represent rests on a calibrated mobility rather than an estimated one.

The proposals that earmark the proceeds, McMorrow's Workforce Reinvestment Fund and the ten per cent lockbox floated by Gabriel Weinberg, are really about the spending side, and the model has two things to say about them. The first is a funding constraint: the transfers these schemes promise are affordable over the transition only if the revenue that fills them reaches the rent, because the income-tax base that would otherwise pay for them is itself shrinking as the wages disappear. The instrument debate and the worker-protection debate are the same debate. The second is that the choice between banking the proceeds and rebating them to citizens is the one genuinely political fork in the whole analysis: banking builds public net worth and brings ownership of the automated economy back home, while rebating delivers lower inequality sooner but no public stake. A fund that finances genuinely new kinds of work, rather than only cushioning lost ones, also acts on the reinstatement margin that governs unemployment, though whether retraining creates new tasks is an input to the model rather than something it derives.

That leaves the cautious institutional positions, and the proposals the model is not equipped to score. The International Monetary Fund's warning that an AI-specific tax could blunt productivity is supported, in the sense that the model does show a real efficiency cost once deployment responds to the levy; but the broader bases the Fund prefers, ordinary corporate tax in particular, reach almost none of the durable surplus in this economy, so the caution cuts both ways. Two proposals fall outside what the model can represent at all, and it is better to say so than to stretch. Elizabeth Warren's excise on data-centre electricity addresses an energy base, and the model has no explicit energy or compute-cost channel, so it cannot evaluate it. The Brookings preference for folding an AI charge into a consumption-side value-added tax, with business-to-business use exempted, is related in spirit to the source levy the model does study, since both tax value where it is recognised and both resist relocation, but the model does not represent a value-added tax with a business exemption, so it cannot compare that design directly. These are the clean edges of what the exercise can claim.

8.4 Tax the rent, not just the machines

The single clearest implication is that the instrument has to reach the AI IP rent, because that is where the durable surplus sits once the economy has automated. A robot tax is administrable and lands cleanly on the physical robotic income, but it taxes the competitive, low-rent channel and misses the rent entirely; the ordinary corporate tax reaches almost nothing, because the rent is a deductible licence fee and the competitive capital return is competed to zero in the mature economy. Only a digital-services-style levy on AI revenue, or a withholding on the rent as it is repatriated, touches the surplus. If the UK wants the automation transition to fund anything, the levy on AI value (in the spirit of the existing digital-services tax, which already taxes revenue precisely because revenue resists shifting)[1] is the load-bearing instrument, and the robot tax is a useful complement rather than the centrepiece. There is one alternative to taxing the rent rather than complementing it, namely competing it away (subsection O): promoting a contestable frontier shrinks the rent and so curbs foreign ownership without a levy, but it leaves nothing to tax, so it serves the ownership and consumer-welfare goal rather than the fiscal one.

8.5 Watch ownership, not only revenue

The deepest result is about who ends up owning the capital stock, not about the tax take. An untaxed, reinvested foreign rent does not merely drain a flow; it lets the foreign owner accumulate domestic equity, climbing toward seventy per cent of the capital stock in the mature economy (Figure 8). For a goal of long-run stability and domestic resilience, that ownership drift is arguably more important than the revenue line, because it determines whether the returns to the automated economy accrue at home or abroad. The instruments that reach the rent also curb the drift, the rebated withholding most of all. A policy that banks the proceeds in a sovereign equity fund (rather than rebating them) does double duty here: it both raises net public saving and re-domesticates ownership, which is the combination the full-toolkit scenario achieves.

One nuance belongs here, because it is easy to assume that buying the ownership back is equivalent to taxing the rent. It is not, and the difference is taxing rights. Acquiring the foreign owner's equity in the automated capital does repatriate the after-tax return on that capital, which addresses the ownership-drift problem directly. But the taxing rights on the AI IP rent itself follow the company's domicile, so the rent is still taxed first abroad, at the owner country's rate, and only the residual after that first slice flows home with the equity. A sovereign fund that buys the equity therefore re-domesticates the *ownership* of the surplus without re-domesticating the *taxing rights* over it; the host concedes the first cut to the rent's home jurisdiction. This is why a source levy and an ownership stake are complements rather than substitutes: the levy reaches the rent at source where it is recognised in the host, while the fund captures the ownership return net of the foreign tax. Reaching the rent and owning the capital are two different goals, served by two different instruments.

[1] The instrument the model recommends is best read as a widened, AI-scoped levy rather than the digital-services tax as it currently exists. Real-world DSTs (the UK's two per cent levy, and the French, Italian and Spanish equivalents) are scoped narrowly to digital advertising, search and online-marketplace revenue, and explicitly exclude cloud computing and the sale of AI or software services. The existing DST is therefore the closest live analogue for how a recognised-revenue base resists relocation, not the base the model is pointing at; reaching the AI rent would require extending the scope to AI revenue specifically.

8.6 If the rent rises, every recommendation matters more

The recommendations here are read off the conservative case in which the rent stays a fixed share of output. The rising-rent results (section 6, Figure 26) say the more likely direction is up, through greater pricing power and through AI-native production capturing a growing share of output, and that this changes the policy calculus in degree rather than in kind. Re-running the instruments against a risen rent confirms the ranking is unchanged: a source levy still reaches it and a robot or corporate tax still misses it, because a larger rent is still a rent. What changes is how much is at stake. Under a rent driven to a quarter of output by pricing power, leaving it untaxed lets foreign ownership run to about eighty per cent of the capital stock, while the digital levy paired with a withholding pulls that back to the low forties and lifts public net worth several-fold, a wider gap than the same instruments open in the flat-rent case. In the extreme, where the rent approaches half of output, the untaxed economy does not merely transfer ownership but strains its public finances to the point of insolvency, whereas the full toolkit keeps it solvent and holds foreign ownership below sixty per cent. The instruments do not change; their value, and the cost of forgoing them, scales with the rent.

Three of the recommendations are promoted by this. Watching ownership rather than only revenue moves from prudent to essential, because a rising rent is precisely the case in which the revenue captured understates the ownership transferred. Intervening early matters more, because the drift the early levy is meant to curb is faster and runs further. And competition policy gains weight as a first-best, because preventing the concentration that lets the markup rise is the one route that stops the rent growing in the first place rather than taxing it after the fact; the open-weight dynamic in subsection O is, in this light, not only an erosion of the rent but a brake on its rise. The banking-versus-rebate choice tilts marginally toward banking, since a near-total ownership drift is the case in which rebuilding a domestic stake is most valuable. The retaliation trade-off discussed below sharpens in the same direction: a larger prize raises both the revenue forgone by folding and the value of holding the line.

8.7 Intervene early, and prefer stock to flow taxes

Two further results bear on design. First, the timing result (section 7) shows the cost of waiting is convex and the window for cheap intervention narrows as automation deepens, so a credible instrument introduced early dominates a larger one introduced late. Second, across the policy frontier (section 6A) the stock taxes (a wealth tax) occupy the favourable region on both inequality and solvency, while pure flow taxes leave inequality near the laissez-faire level; a behavioural wealth tax still redistributes at realistic rates even after its base erosion and the modest capital flight it induces. For contained inequality, a wealth-type instrument is therefore worth more than its headline rate suggests, provided it is paired with the rent-reaching levies above so that the automated surplus is itself in the tax base.

8.8 Sovereign compute helps, but is not the lever that reaches the rent

Onshoring the AI servers (raising s_home) is worth doing for resilience and for the transition tax base, when capital returns are briefly high, but it barely moves the steady state on its own, because the durable surplus is the mobile IP rent rather than the competitive return on the servers. The reading is that a sovereign-compute push and a levy on the rent are complements, not alternatives: compute policy widens the base that the levy then taxes, and keeps the physical infrastructure (and its resilience) at home, but it does not substitute for an instrument that reaches the rent wherever the servers sit.

8.9 The binding constraint in practice: digital taxes and retaliation

One thing the model deliberately leaves out has to be put back for the policy to be realistic. The source levy it recommends is, in the real world, a digital-services tax or a withholding, and the United States treats these as discriminatory taxes on its firms and retaliates against them. The instruments are live and contested: the US ran Section 301 tariff investigations into these digital taxes from 2019 to 2021 (covering France, Austria, Italy, Spain, Turkey and the United Kingdom, with twenty-five per cent tariffs determined and then suspended pending the OECD process), and in early 2025 a presidential memorandum directed their renewal while Congress drafted a retaliatory tax (the proposed Section 899) aimed at countries with such measures. The pressure has largely worked: Canada repealed its digital tax, India dropped its equalisation levy, Italy agreed to reform, and the United Kingdom's measure has been reported as a likely concession, while France, pushing to raise its rate, has been warned to expect proportionate retaliation. So the model's result, that a rent-importing host technically can reach the surplus and that server relocation does not defeat it, comes with a political price tag the model does not carry: using the instrument means accepting the risk of trade retaliation. It is worth saying why this fight may be worth having when the earlier digital-tax fights, fought over a couple of points of advertising and marketplace revenue, often were not. The AI case is higher-stakes on two margins at once: the rent that leaks abroad is far larger and compounds into ownership of the capital stock, and the wage base that funds the state thins at the same time as the displaced roles stop being taxed. The original DST disputes were about a slice of revenue; this one is about the ownership of the automated economy and the integrity of the fiscal base together, which is why a host has more reason to absorb the retaliation here than it did over the first generation of digital taxes.

This sharpens the recommendation rather than overturning it, and it makes the timing argument concrete. The prize at stake grows as automation deepens, because the rent compounds into ownership and capital's share of income rises, so the value a country forgoes by folding to avoid retaliation increases over time. A country that wants to keep the rent and slow the loss of ownership by taxing it will, on current evidence, have to be willing to absorb retaliation to do so, and the case for absorbing it strengthens the further automation proceeds. The competition result above is the important escape from the bind: backing open-weight competition erodes the rent without a levy, and so without provoking retaliation, at the cost of collecting no revenue. The blunt choice for a rent-importer is therefore three-way, not two: tax the rent and accept the retaliation, foster competition and forgo the revenue, or fold and accept the loss of ownership.

8.10 A coherent package, with one explicit choice left to politics

Putting these together, the model's conditional recommendation for the stated goals is a package rather than a single tax: a digital levy on AI value as the centrepiece (reaching the rent), a robot tax as a clean complement on the physical channel, an onshored-compute policy to widen the transition base and keep infrastructure at home, a behavioural wealth tax to hold inequality down, and a commoditisation lever, backing open-weight models and a contestable frontier so that no durable monopoly forms in the first place, with the proceeds of the levies banked in a sovereign fund rather than fully rebated. The commoditisation lever sits a little apart from the others, because it works by shrinking the rent rather than capturing it (subsection O): it serves the ownership and consumer-price goal and provokes no retaliation, but it leaves nothing in the public purse, so it complements the levy rather than replacing it, and it is the one lever a rent-importing host can pull that a host of the providers would resist, since it would be commoditising its own champions. The instruments are complements: in the experiments the full toolkit raises the most revenue and holds foreign ownership to about half the stock. The one genuine choice the model cannot make is distributional: a rebated withholding delivers the lowest inequality but builds no public saving, while a banked levy builds the fund and re-domesticates ownership but does less for measured inequality in the near term. That trade between paying citizens now and accumulating a public stake is a political judgement about whether the state or the household should hold the claim on the automated economy, and the model can only lay out the consequences of each, not choose between them.

8.11 Did the experiments change the recommendation?

It is worth being explicit about how much of this recommendation was already in place after the core two-channel experiments (subsections F to I) and how much the later extensions moved it. The short answer is that the centrepiece did not change, the case for it was hardened against its strongest objections, and one genuine rider was added, with the menu of complements widened around it.

What did not change. The load-bearing recommendation, a source levy that reaches the AI rent, a focus on ownership and not only revenue, and the bank-versus-rebate distributional choice, was set by the two-channel and open-economy experiments and survived every later test intact. None of the extensions displaced the digital levy as the centrepiece or overturned the ownership story.

What hardened it. The debate tests (subsection K) put the two strongest objections directly to the model. Friedman's claim that the base self-destructs the moment it is taxed came out as costly but not self-defeating in the credible range; the territorial objection came out as decisive for the United States but not for a rent-importing host. The capitalisation result (subsection P) reinforced the same conclusion from the asset side: the surplus is foreign-held intellectual property that a domestic wealth tax cannot reach, while a source levy lowers its present value directly. These did not change the recommendation, they removed the main reasons to doubt it.

The one genuine change. The optimising-owner extension (subsection N) qualified the territorial result. The earlier finding that reach does not depend on where the compute sits is correct, but the owner can still move where the rent is *booked*. This adds a rider that was not present at the two-channel stage: the levy has to be written against recognised domestic value and paired with transfer-pricing and anti-avoidance rules, or the base leaks through the accounts even though it cannot leak through the data-centre map. This is the clearest case of an experiment changing the policy, from "the base cannot move" to "the base moves through recognition, so that channel has to be closed too".

What widened the menu. The contestable-frontier result (subsection O) added competition policy as a genuine alternative route to the ownership goal, first-best if the aim is that the rent should not exist, but raising no revenue, so it complements rather than replaces the levy. The unemployment and safety-net result (subsection L) tied the worker-protection debate to the instrument debate by showing the transfers the proposals call for are affordable only from a rent-reaching levy. Neither displaced the centrepiece; both attached new commitments to it. The net effect is a recommendation that is more robust and slightly more demanding than at the two-channel stage:

the same digital levy, now explicitly paired with anti-avoidance rules, with competition policy offered as an ownership-only alternative, and with the safety net's funding shown to rest on the very same instrument.

What raised the stakes. The rising-rent results (Figure 26) did not change the instrument ranking, which holds because a larger rent is still a rent reached by the same source levy. What they changed is the magnitude of the problem and the urgency of the response: a rent that grows with automation drives the ownership drift toward the whole capital stock and, untaxed in the extreme, toward insolvency, so the ownership focus and the early-action argument are not merely supported but made central. This is the sense in which the flat-rent results elsewhere are a floor; the recommendation is the same, but the consequence of ignoring it is larger, and competition policy gains a second rationale as the route that keeps the rent from rising at all.

9 Summary of findings

Finding	Status in v3	Detail
Stock taxes vs flow taxes	Holds, refined	Stock taxes still occupy the favourable region; flow taxes leave inequality at the laissez-faire level.
Wealth-tax "dominance"	Reframed	Still redistributes at realistic rates; a modest, rate-increasing share of apparent equality is capital flight.
Composition effect	Corrected	Measured vs true Gini wedge stays under a point of Gini at all rates; compression is front-loaded; offshore share scales ~2% per pp.
Concentration engine	Microfounded	Tail emerges from heterogeneous persistent returns, not an imposed kernel.
Robustness (Sobol)	New	Inequality driven by the dispersion and persistence of returns, not the behavioural elasticities; fiscal results driven by the CES elasticity.
Fiscal solvency (H1)	Supported under this closure	Real economy is locally stable in every regime (capital-map slope < 1); solvency turns on the $r>g$ debt root, which under-taxed welfare states (15% income tax + UBI) hit and adequately-taxed regimes neutralise. This is a numerical result under the residual-investment, full-employment closure, not a general theorem.
Transition realism	Fixed	Differential-saving closure gives a bounded, convergent path (no explosion); K/Y rises from ~2.9 to ~8–10 as the economy automates and settles.
Two-channel robustness	New	The ranking of instruments holds across the rent size (Figure 13); headline revenue and ownership are tight across seeds (Figure 14).

Finding	Status in v3	Detail
Rising rent	New	If pricing power grows or the AI cluster captures a rising output share, the rent climbs from about an eighth of output toward a quarter or more and foreign ownership accelerates toward the whole capital stock (Figure 26). The instrument ranking is unchanged, but the stakes and the urgency of reaching the rent rise sharply; the flat-rent results elsewhere are a conservative floor.
Robot-IP rent	New	If robots carry their own foreign-held IP rent (the model running them) rather than being purely competitive hardware, the total rent roughly doubles and ownership drifts toward the whole stock (Figure 27). A hardware robot tax misses it; the same source levy reaches it; the ranking is unchanged. The meaningful distinction is competitive hardware versus rent-bearing IP, not robots versus AI, so confining the rent to AI is conservative.
Open-economy ownership	New	With a current-account leak the ownership drift is a range, from ~70% (closed) to 0% (full repatriation), and repatriation cuts domestic output to ~70–75% (Figure 15).
Cost of taxing the rent	New	With an AI-supply response the levy is no longer free: output falls to ~78% of untaxed while the revenue share is broadly unchanged (Figure 16).
Reinstatement and the labour share	New	New tasks make the labour-share fall conditional, not mechanical: it recovers from ~40% to ~60% as reinstatement strengthens, at some cost to output via lower saving (Figure 17).
Unemployment and the safety net	New	Extensive-margin displacement gives a 10–20% unemployment rate under balanced reinstatement; the safety net is affordable only if funded from a rent-reaching levy (Figure 20).
Optimising foreign owner	New	An owner that shifts where the rent is booked bounds the host's take (it bends below the non-strategic line) and drifts ownership up; reach survives server relocation but not profit-shifting of the recognition (Figure 22).
Contestable frontier	New	Competition erodes the markup: it cuts foreign ownership like a tax does, but destroys the rent rather than capturing it, so it is a substitute for the ownership goal and the opposite for revenue (Figure 23).

Finding	Status in v3	Detail
Capitalised rent (IP value)	New	The rent capitalised as a stock is worth about 2.2 years of output and lifts foreign ownership at market value above book; a source levy erodes it, while a market price on domestic equity is degenerate because the surplus is the foreign-held IP, not the domestic capital (Figure 24).

10 Caveats and remaining work

The model's claims are bounded by its assumptions. The limitations below are ordered by how much each one constrains what the results can claim, with the first group the priorities for the next iteration. They concern the specific results; the foundational assumptions of the modelling approach itself, and which conclusions are robust to them, are catalogued separately in Appendix A.

10.1 What most bounds the headline claims

- **Open-economy closure.** *Partially addressed.* A parametric current-account leak now lets the foreign owner repatriate a share of its after-tax rent as real goods rather than reinvesting it, so the ownership drift in Figure 8 is reported as a range between the closed pole and full repatriation (Figure 15) rather than as a single upper bound. What remains is that the repatriation share is a parameter, not an estimated behavioural choice: there is still no exchange rate, no relative-price adjustment, and no optimising foreign owner deciding how much to repatriate, so the range is bounded correctly but its interior point is not yet pinned down. This stays the largest structural caveat on the most striking result.
- **The AI rent: its size and its incentive effects.** *Partially addressed.* The ranking of instruments is now shown to be robust to the rent size across a markup sweep (Figure 13), so the qualitative conclusions no longer hinge on the calibrated mu_frac, and an AI-supply elasticity to the net-of-tax rent now gives the levy an efficiency cost, so taxing the rent is no longer free (Figure 16). Two things still bound this: the markup and the supply elasticity are now anchored to plausible central values (mature-FDI reinvestment shares for the repatriation rate, the -0.5 to -1.0 user-cost elasticity for the supply response, see the calibration note in section 6I) but are not estimated for this specific market, which has no tax-variation history to estimate from; and the deadweight loss is capped per period because the model's accumulation channel amplifies any persistent output loss, so the magnitude of the cost is indicative rather than calibrated. The model can now claim a net rather than gross benefit in direction, and a disciplined range, but not yet a point estimate of the cost.
- **Task creation and unemployment.** *Addressed, with calibration caveats.* Both halves of the labour-market extension are now in: a reinstatement margin makes the labour-share fall conditional rather than mechanical (Figure 17), and an extensive-margin overlay turns part of that fall into measured unemployment with a funded safety net (Figure 20), so the model can speak to the displacement debate directly. The unemployment is structural (workers displaced by capital, output held) rather than cyclical, which suits an automation model but means it does not capture demand-driven joblessness. The remaining limits are calibration, not

structure: the displacement pass-through, the benefit replacement rate and the UBI labour-supply response are anchored to literature ranges (the calibration table in section 6M) but not estimated for this market, and the sensitivity analysis (Figure 21) shows the unemployment rate is the most parameter-sensitive outcome in the model, swinging on the pass-through, so it is reported as a range set by the displacement-versus-reinstatement balance rather than a point.

- **The wage result depends on labour scarcity.** *Addressed (experiment R), and a required qualification on a stated result.* The headline model pays each labour type its marginal product in fixed supply, so the task automation cannot reach becomes a scarce complement and its wage rises in level even as the labour share falls. That level claim is not general: it holds *only* because supply is fixed. If the remaining human task is one any worker can do, abundant labour competes the premium away and the surplus accrues to capital instead (experiment R measures this: the routine/cognitive wage premium collapses from about thirty-six to about three and the bottleneck cluster's labour share falls). If instead it is a genuinely scarce, hard-to-acquire skill, the premium survives but is captured by a narrow wage elite rather than by labour broadly. The reassuring reading of the headline result is therefore the knife-edge case; the two generic outcomes both worsen the distribution, so the caveat sharpens the paper's conclusion rather than threatening it. What remains is to give labour an explicit participation and skill-heterogeneity margin so the three cases can be weighted rather than bracketed.

10.2 What would sharpen the results

- **AI as a producer, not only a priced input.** The model treats AI as a factor inside one representative firm that pays a markup, and the rising-rent subsection lets that markup and the cognitive cluster's output share grow in reduced form. It does not yet model competition between distinct AI-native and human-incumbent firms, in which an AI-native producer captures the entire output of the firms it displaces rather than a markup on a cluster. A heterogeneous-firm version, with the owner of the AI-native firm either domestic or foreign, is the natural way to model the difference between AI used inside a company and AI used as the company, and is the most consequential remaining structural extension.
- **Endogenous markup.** Addressed (subsection O): the markup now falls with market contestability, so competition policy can be set against taxation as a route to the rent. What would further sharpen it is making entry respond endogenously to the size of the rent, rather than treating contestability as an exogenous state.
- **A market-clearing equity price.** Attempted (subsection P), with a negative result: a market price on domestic equity is degenerate, because the competitive return is competed away and the durable surplus is the foreign-held IP, so the value sits in the capitalised rent the reduced-form valuation already reports. A genuine asset-price and bidding-up dynamic would need the rent-bearing firm restructured as a traded asset with competing buyers, which remains future work.
- **An explicit welfare criterion.** Policy comparisons remain positive (inequality and solvency), not welfare-optimal; a social welfare function would move the policy section from directional rankings to normative recommendations and let any "preferable regime" language be stated properly.
- **An optimising foreign owner.** Addressed (subsection N): the owner now shifts where the rent is recognised in response to the host's wedge, which bounds the territorial reach and makes transfer pricing the binding margin. What would further sharpen it is endogenising the

compute-location and repatriation choices as well, and anchoring the profit-shifting elasticity to estimates rather than a calibrated value.

- **A stacked compute rent.** Implemented as experiment Q (subsection Q): a second foreign rent (`mu_compute`) on the compute layer, embedded in the AI capital-good price rather than a licence flow, with its own domicile and reachability. It confirms that the compute rent accelerates ownership drift and is missed by the digital levy and the withholding, reached only by a tariff, a usage levy, or a domestic chip-maker. What remains for future work is to give the two rents independent persistence (the contestability dial of experiment O asked of the compute layer, to test the durability asymmetry between a sticky model rent and a possibly-commoditising hardware rent), and the more literal up-front capex form of the markup rather than the recurring wedge used here.
- **Elastic labour supply and endogenous automation.** Implemented as experiments R and S. R relaxes fixed labour supply and shows the bottleneck-wage spike is an artefact of inelastic supply (the scarcity premium competes away), though the aggregate labour share need not fall. S makes automation cost-driven rather than an exogenous ramp and shows the transition still arrives but at a pace set by the cost decline. Both remain reduced-form: a richer treatment would give labour an explicit participation margin and an extensive unemployment response in R, and a calibrated per-task cost distribution (rather than a single threshold) in S.

10.3 Realism and robustness

- **International-tax architecture.** The withholding sits above typical treaty caps, and the model ignores treaties, Pillar Two, and the retaliation a unilateral digital levy invites (the actual US/China flashpoint); constraining instruments to treaty-feasible ranges and adding a strategic dimension would make the policy claims implementable.
- **Empirical validation.** The two-channel results are internally consistent (the stock-flow invariants and the production Euler condition hold to machine precision) but are not yet matched to external moments (observed factor shares, FDI stocks, AI-sector margins); calibrating to these would turn the magnitudes from illustrative into estimated.
- **Credit, leverage, housing and asset prices.** There is still no credit or leverage and no housing or asset-price sector; the former is where the closest prior art locates instability, the latter where much of the empirical capital-share movement actually lives.
- **Behavioural realism elsewhere.** The UBI is a transfer with no labour-supply response, which the experimental evidence suggests is first-order for welfare; the behavioural elasticities are calibrated to the literature but are not structural (the Sobol analysis in section 6E reports how much the results lean on them); and the two-channel headline numbers are now reported with seed dispersion (Figure 14), which is small.

Declaration of generative AI use

During the preparation of this work the author used large language models, Claude Opus 4.8 (Anthropic) and GPT-5.5 (OpenAI), as coding and writing assistants: to help implement, debug, and refactor the simulation and analysis code and to produce figures, and to draft and edit the manuscript. The author reviewed and edited all such output and takes full responsibility for the model, the experiments, the results, and the conclusions presented here.

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A Appendix A: Assumptions and limitations

This appendix catalogues what the model assumes and where it is most likely to mislead. Section 10 lists caveats specific to individual results and the work that would sharpen them; this appendix is the foundational list, the assumptions built into the modelling approach itself. The short version: treat the structural orderings as reliable and the magnitudes as illustrative, and pair the model with the things it cannot see.

A.1 A.1 What kind of model this is

Stock-flow-consistent and agent-based models of this kind are consistency engines, not forecasting machines (the Godley and Lavoie tradition). Their value is that money cannot appear or vanish, every assumption is explicit and can be swept, and the channel driving each result can be switched on and off. They answer conditional questions ("if the rent is mobile and foreign-held, which tax reaches it?") well, and unconditional questions ("what will ownership be in year X?") badly. Box's aphorism is literal here: the model is wrong, and it is useful for orderings and mechanisms rather than point estimates.

A.2 A.2 The central assumption: that the rent exists

The entire analysis rests on a large, durable, separable, foreign-held AI rent. This is calibrated, not estimated. The frontier labs lose money today, so the rent is a bet on a future durable margin, and our own competition result shows it can erode if the frontier commoditises. It is also not cleanly observable: separating an IP rent from a competitive return and from ordinary profit is a contested measurement problem, so the surplus the model treats as a clean markup is, in practice, reached only through proxies (a revenue or consumption base, a withholding on the licence fee, or a residual-profit allocation such as Pillar One's Amount A) and adversarial transfer-pricing rules. If the rent is smaller, more competed, or harder to characterise than assumed, the problem the paper addresses shrinks with it.

A related assumption is that the rent sits only on the AI side. The model treats the robotic channel as purely competitive (reproducible hardware earning a normal return), so all the rent is on the cognitive cluster. This is conservative, because a robot is hardware plus IP, and the IP that runs it (the control or foundation model, plus the proprietary design) is scarce and foreign-held in the same way the AI model is. An off-default lever (robot_ip) puts an embodied-IP rent on the robotic cluster and confirms it behaves exactly like the AI rent: it raises the total foreign-held rent (from about an eighth of output to about a quarter at a robot markup equal to the AI one), accelerates the ownership drift, and is reached by the same source levy while a hardware robot tax misses it, leaving the instrument ranking unchanged (section 6, Figure 27). The economically meaningful distinction is therefore competitive hardware versus rent-bearing IP, not robots versus AI; confining the rent to the AI cluster understates rather than overstates it.

A second qualification concerns the *form* of that hardware-side rent, and it is less comfortable. The robot_ip lever still treats the embodied rent as a flow reachable like the AI rent, but the compute layer has both a distinct rentier and a distinct form. The scarce asset there is the accelerator design and its software ecosystem (the NVIDIA-and-CUDA lock-in), whose margin is largely foreign-held and, unlike the model's recurring licence fee, embedded in a capital-good price paid up front. That difference bites on the headline instrument: a withholding or digital-services levy sits in the path of a cross-border licence flow but not in the path of a goods price, so a compute-embedded chip rent can escape the very source levy this paper recommends, and is reachable only by a tariff or customs charge on imported compute, or partially by a usage levy if the provider passes it through. Disaggregating the single foreign owner into a chip-maker and a model-owner, each with its own size, durability, location and taxability, would therefore deepen the importer's predicament (more foreign rent, faster ownership drift) while adding one rent that even the reaching instruments miss. We implement this as experiment Q (subsection Q), which confirms both halves: a stacked compute rent raises foreign ownership, and the digital levy and withholding collect nothing from it while a tariff or usage levy do. The point to register here is that confining the rent to a single, withholding-reachable *form*, like confining it to the AI cluster, understates rather than overstates the problem.

A.3 A.3 Production and technology

Output is an aggregate nested CES function with a small number of substitution elasticities. Whether automation raises or lowers the labour share turns on whether the relevant elasticity exceeds one, which is empirically contested and may not be a stable structural object at all (the Cambridge-capital objection that aggregate capital is not always coherent). The task framework (Acemoglu and Restrepo) is the modern repair and supplies the reinstatement margin, but Rognlie's caution that much of the measured capital-share rise is housing and markup rather than productive capital is a live objection. Crucially, the automation path is imposed as an exogenous logistic ramp: the model gives the consequences if AI automates cognitive work on that schedule, not whether or how far it will. The largest real-world uncertainty is an input, not a result.

A.4 A.4 Closure and finance

Several closure choices bite directly on the headline numbers. Output sits at potential, so there is no demand-side recession from displacement, arguably the outcome the displacement literature fears most. Saving equals investment by construction, equity is valued at book, and there is no inflation, monetary policy, banking, credit cycle or asset market, so financial fragility and bubbles are absent. Most consequentially, the baseline is a closed economy, so the foreign owner can only recycle its rent into domestic assets, which is what mechanically produces the dramatic "buys the whole capital stock" result; adding a trade balance turns that single number into a range. The ownership headline should therefore be read as the fully-reinvested pole, not a central estimate.

A.5 A.5 Behaviour and expectations

Agents follow fixed behavioural rules rather than re-optimising when policy changes, which invites the Lucas critique: in reality the powerful party, the foreign rent-owner, would respond hardest, and most of that response is unmodelled (the one channel we did open, the owner choosing where to book the rent, mattered and forced the anti-avoidance rider). There is also no genuine uncertainty or expectations formation, so there is no precautionary behaviour, no investment boom on anticipated AI returns, and no front-running of a pre-announced tax. Beliefs about where AI is heading, a dominant force in the real story, are absent.

A.6 A.6 Aggregation: who loses is invisible

The economy has one good, no input-output structure across sectors, a coarse cognitive-versus-routine labour split, no spatial or regional dimension, and (in the baseline) a single representative firm rather than competing AI-native and incumbent firms. AI's impact is highly uneven across industries, regions and skills, and none of that texture survives. The model speaks to the labour share and the wealth distribution in aggregate, not to which workers or places are hit, and the rising-rent output-capture channel is a reduced-form proxy for firm-level displacement rather than a mechanism (see section 6 and section 10).

A.7 A.7 The distribution engine

Wealth concentration is generated by a heterogeneous-returns random-growth process (Benhabib, Bisin and Zhu; Fagereng and co-authors). The inequality results lean on the assumed level and dispersion of returns to wealth, which is among the least-settled inputs empirically and one of the main drivers of the headline distributional numbers. The finding that stock taxes compress inequality while flow taxes do not is partly a property of how this engine is built, and should be read as internally consistent rather than independently established.

A.8 A.8 What the model is silent on

The silences are where a policymaker is most exposed, because they never appear as caveats inside the model. It assumes the levy can be imposed and collected, but the binding real-world constraint is political economy: the owner lobbies, threatens to withdraw, and its home government retaliates, which the paper treats as narrative rather than as a modelled strategic game. It focuses on the distribution of the surplus and largely ignores the consumer-surplus gains if AI makes goods cheaper, so it may understate the size-of-pie benefits. And it has no welfare function, so any "preferable regime" language is the authors' gloss, not the model's output.

A.9 A.9 The deepest limitation: a moving structure

Structural modelling assumes deep parameters (elasticities, propensities) that survive the policy change being studied. The sharper worry here is not the Lucas critique narrowly but that the entire subject is a regime change, in what a firm is, how value is produced, and what labour does, modelled with parameters calibrated to the world before that change. If AI genuinely transforms production, the structure itself is moving, and a model fitted to the recent past has uncertain ground beneath it. This is the strongest single reason to read the work as disciplined conditional reasoning rather than prediction.

A.10 A.10 What to trust, and what to discount

Putting the above together, the conclusions sort into two groups.

- **Robust (structural, close to accounting facts).** Which base reaches the rent and why (a source levy over a robot, corporate or token tax); the territoriality asymmetry (decisive for the provider's home, weak for a rent-importing host); and the direction that an untaxed, reinvested, foreign-held rent transfers ownership over time. These depend on the sign and presence of a mechanism, not on magnitudes, and survive sweeping the uncertain parameters.
- **Fragile (calibrated, assumption-sensitive).** Every magnitude: the rent at an eighth of output, the seventy per cent ownership share, the rising-rent quarter or half, and above all the unemployment range, which the sensitivity analysis flags as the least anchored output. These should be read as illustrative ranges set by parameters that are bounded but not estimated.
- **Outside the model.** Enforceability and political economy, demand-side macro, the pace and reach of AI capability, and welfare. A recommendation can be correct inside the model and still fail in the world for these reasons, so the model is a complement to judgement on them, not a substitute.

The intended use follows from this split: base the recommendations on the orderings and the framework, report the numbers as ranges, and weigh them against the constraints the model cannot see.